SF LIVE SUITE

REFERENCE MANUAL

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31 PLUGINS FOR PERFORMING HYPNOTIC / DUB TECHNO IN ABLETON LIVE
10 INSTRUMENTS 12 EFFECTS 9 MASTER AND MIX TOOLS

VERSION 0.2 NOVEMBER 2026
STRANGE FREQUENCIES

CONTROL RANGES AND DEFAULTS IN THIS MANUAL ARE GENERATED FROM THE PLUGIN CODE.

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1 HOW THE SUITE WORKS

The suite is built for one kind of performance: very simple note content, often one note in a pattern that never changes, with all the variation played on knobs. Everything follows from that. Every plugin is designed to be played, not programmed, and every plugin talks to the others.

THE RIG

One Live set that never changes. The track layout is fixed: one track per role (kick, percussion, rumble, bass, lead, voice, pad, deck). The OXI One MkII sequences everything; the OXI E16 plays the knobs. Songs are not projects: a song is a stored set of values in every plugin, recalled on the bar. You switch songs without loading anything.

THE PERFORMANCE ROW

The first eight controls of every plugin are its performance row: the large knobs at the top of its window, and the first eight parameters Live shows for the device, so one E16 row maps one plugin. Everything below the display is set-and-forget: models, ranges, routing, behaviour.

DENSITY, GATE, FILTER

Three controls cover most of what changes in a track, and they appear wherever they make sense. **DENSITY** is how many events happen (ghost notes fill the gaps). **GATE** is how long each event lasts. **FILTER** is what each event sounds like: a bipolar DJ filter, centre open, left low-pass, right high-pass. Play two against each other: density up with gate down is a tight, nervous build; density down with gate open is spacious dub; density up with the filter closed is tension waiting to be released.

SYSTEM CONTROLS ON EVERY PLUGIN

Below every plugin's controls: the MOTION row (two modulators) and the SYSTEM row (SLOW, MACRO). They are the same on every plugin and are not repeated in the module tables.

CONTROL	RANGE	WHAT IT DOES
SLOW	CONTROL, 1-64 BAR, OFF	Slow mode. Every performance-row knob travels to where you set it over exactly this many bars, however far you moved it; song recalls glide too. CONTROL (the default): SF CONTROL's SLOW ALL sets the length. A move keeps the length it set off with. While travelling the knob shows GLIDE now>target in green.
MACRO SRC	OFF, MACRO 1-6	Which SF CONTROL macro this plugin follows.
MACRO AMT	-100 to +100 %	How strongly and in which direction. Each plugin decides which of its knobs respond (the ones marked M in the tables).
M1 / M2 SOURCE	OFF, SINE, TRIANGLE, SQUARE, RAMP UP, RAMP DOWN, RANDOM, DRIFT, KICK, NOTES or INPUT	Two modulators (MOTION). Waves, RANDOM (stepped) and DRIFT (smooth) move around the knob; ramps rise or fall from it and restart every RATE; KICK follows SF KICK's hits; NOTES (instruments) follows velocity; INPUT (effects) follows the input level.
M1 / M2 RATE	1/16 to 64 BAR, odd	Cycle length, locked to Live's song position; the odd ones run against the bar (MOTION, below). KICK, NOTES, INPUT: the release.
M1 / M2 TARGET	OFF or any knob	The control the modulator moves.
M1 / M2 DEPTH	-100 to +100 %	How far, as a share of the target's whole range. Negative turns it upside down.

MOTION

Two modulators in every plugin, M1 and M2. Pick a SOURCE, a RATE, the TARGET knob and a DEPTH. Everything is locked to Live's song position: every plugin's modulators move together, a 64-bar ramp lands on the 64th bar, and a set replays the same way every time, RANDOM and DRIFT included (their seeds are saved). Modulation adds to the knob, the macro and SLOW; a bar on the knob shows where it is. Steps (SQUARE, RANDOM, a ramp restart) are smoothed over 3 ms. MOTION is part of songs and presets.

— **Hypnotic movement:** DRIFT on a filter at 8 or 16 BAR, DEPTH 10-20 %: it never repeats, so neither does the loop.

— **Long builds:** RAMP UP at 32 or 64 BAR on ENERGY or a filter; turn DEPTH to set how far it goes.

— **Polyrhythm:** the odd RATEs (3/16, 5/16, 7/16, 1/4D, 1/2D, 1/4T, 1/2T, 3, 5, 7 BAR) turn against the 4/4: a 5/16 SINE on a filter meets the bar again every 5 bars.

— **Kick-driven:** KICK on a bass's DRIVE or a pad's CUTOFF: every kick pushes it, RATE sets the fall.

— **Played:** NOTES on an instrument (velocity) or INPUT on an effect (level) makes the sound respond to how hard it is played.

KEY

Write parts in C. SF CONTROL's KEY (stored per song) then plays them in the song's key on every SF instrument: BASS, LEAD, PAD and VOICE transpose by the nearest interval (-6 to +5 semitones); KICK, RUMBLE and STRIKE land their own pitch on the key root, so the low end is always in tune with the track. Notes already sounding follow a KEY change. KEY FOLLOW on each instrument opts out. OFF changes nothing, and without SF CONTROL running every instrument plays in C.

Mind the low register. KEY moves parts up to 6 semitones either way, and KICK, RUMBLE and STRIKE snap to the nearest root, so the low end can land high or very low: a kick body at 50 Hz becomes 69 Hz in C# and 37 Hz in D. Check every song's KEY on a sub. TUNE (stored per song) moves the kick from the snapped root: -12 takes the 69 Hz kick an octave down, +7 or -5 puts it on the fifth. A bass too deep: move the part up an octave on the OXI.

NOTHING CAN HURT THE PA

Every knob is smoothed, so 7-bit MIDI never steps audibly. Resonance is level-compensated. Feedback above 100 % saturates instead of exploding. Every output is clamped at +12 dBFS. A plugin that produces an invalid value, or sits at the clamp for 250 ms (a runaway), outputs silence, resets itself, shows **FAULT** in red and carries on. **PANIC** on SF CONTROL fades every SF plugin out over 5 ms and clears every voice, tail and delay line: for a runaway loop, a stuck note or a wrong recall. While held, everything stays silent (no input, no new notes; DECK keeps its place; every header shows PANIC in red); let go and it fades back in over 5 ms. A set never reopens with PANIC held. Grab any knob in the dark.

THE COLOUR CODE

The plugins are black, white and grey. Colour only carries information:

	COLOUR	MEANS
	RED	Clip, FAULT (the plugin reset itself), missing file, true peak at the ceiling.
	AMBER	Conflict or warning: an ALLOC conflict, a deck armed and waiting, RETURN KNOB on BUILD.
	GREEN	Active or locked: bus connected, kick reporting, a song recall or a drop waiting for the bar, a value gliding (SLOW).
	CYAN	The value of the control you are touching.

2 SONGS AND PRESETS

Every SF plugin stores 16 song slots of its own settings, saved with the Live set. SF CONTROL chooses the song; every plugin recalls its slot on the same bar line while Live plays (at once when stopped).

SAVING A SONG

— Set every plugin how the song should start.

— Press **STORE ALL** on SF CONTROL. Every SF plugin stores its current values into the song in force (SONG).

— **Into another song:** choose it in **STORE TO** on SF CONTROL, then press STORE ALL. That song isn't recalled first, so nothing you hear changes; it becomes the song in force (SONG follows) and STORE TO goes back to ACTIVE SONG. Choosing the song on SONG instead would recall it, and what you had set would be gone before you stored it.

— If a song change is waiting for its bar (WAIT BAR) when you STORE TO, the change still lands and stays the song in force.

— STORE in any plugin's header stores just that plugin into the selected song.

RECALLING A SONG

— Send Program Change on MIDI channel 16 to SF CONTROL (program 1 = song 01). Or turn SF CONTROL's SONG.

— SF CONTROL decides the bar line and sends it with the song (and the song's KEY), so every plugin changes on the same bar whatever order Live processes the tracks in. A change sent within 30 ms after a bar line (on the downbeat) counts for that bar; later in the bar, for the next one. Inside an Arrangement loop that never reaches that bar, every plugin changes on the next bar line after the loop jumps back.

— The header of every plugin shows SONG nn, and WAIT BAR in green while the recall is waiting for the bar line.

— Recalled values are reported to Live, so the E16 rings jump to the new song's settings. A recalled knob is yours again the moment you turn it.

— A plugin with nothing stored in that song stays as it is.

— With SLOW ALL (SF CONTROL) or a plugin's own SLOW on, the recall glides instead of jumping: a song change becomes an 8- or 16-bar transition.

— Arrangement automation wins over recalled values, as on any device.

VST3 AND PROGRAM CHANGE

VST3 plugins don't receive MIDI Program Change as MIDI. SF CONTROL therefore lists 16 programs (SONG 01-16); Live 12.2 and later turn a Program Change on the track into a program change for the VST3, older versions don't. The AU version reads Program Change directly from MIDI in any version. If song changes don't arrive in the VST3, use the AU.

PRESETS

— The **PRESET** button in every header opens the menu: factory presets grouped by style (DUB, ACID, HYPNOTIC ...), your own presets, SAVE USER PRESET and SHOW PRESET FOLDER. The button shows the preset that is loaded.

— A preset sets every sound control of the plugin, MOTION included, and Live is told about every change, so the E16 rings follow. It never touches SLOW, MACRO, ALLOC, ROLE, a source setting or KEY FOLLOW.

— 191 factory presets across 26 plugins. Use them as starting points, then store the result into a song.

— User presets are files in Music/SF Live Suite/Presets/<PLUGIN>. Copy the folder to take them to another computer.

3 POCKET: KICK-AWARE DUCKING

SF KICK reports every hit to the other SF plugins: when it hit, how hard, its pitch sweep, how long it rings and how its filter is set. Any module with Pocket carves a kick-shaped hole in itself. No sidechain routing.

TWO KNOBS

DEPTH (or PUMP, DUCK) sets how deep. **SHAPE** is one knob from transparent to pumping:

— **Left:** a narrow cut that follows the kick's pitch sweep down to its fundamental. The bass keeps its body; you feel more space without hearing a duck.

— **Middle:** the low band (below about three times the kick's fundamental) ducks for the kick's length.

— **Right:** full-band pump, release tied to the tempo.

IT FOLLOWS THE KICK

— **Velocity:** ghost kicks get shallow pockets.

— **DECAY:** lengthen the kick and every pocket lengthens.

— **Pitch:** the cut follows the sweep wherever TUNE and PUNCH put it.

— **Kick FILTER:** high-pass the kick and the low pockets close: the bass and rumble take over the sub. Bring the kick's low end back on the drop and the pockets open on the downbeat.

— **Rolls:** on transparent settings fast hits merge into one dip; on pumping settings each hit pumps.

— **Safety:** if kick tails overlap so much that the pocket would never let go, it shrinks to the attack.

— **Breakdowns:** when the kick stops, pockets close; when it returns they fade back in over a beat.

WHO HAS POCKET

BASS, LEAD, PAD, VOICE (POCKET DEPTH / SHAPE), RUMBLE (PUMP), DELAY and REVERB (DUCK, on the wet signal only), DUCK (on anything), TAP (on any track). SF DUCK can also follow a sidechain input instead of the kick.

TIMING

Pocket looks 512 samples ahead (about 11 ms at 48 kHz) so a kick report is always there before it's needed. Live compensates the delay. Keep Live's buffer size at 512 or less. If the pump lands early or late in your setup, trim POCKET OFFSET on SF CONTROL (+/-20 ms) once, by ear.

AFTER A DROP

SF BUILD's DROP DUCK (default 6 dB) makes every Pocket stage dip under each kick for DUCK TIME after the drop (default 1 bar), whatever its DEPTH, so the first kicks of the drop land on top of the parts and the tail coming back. It follows the kick: back within half a beat of each hit, and easing out over the second half of DUCK TIME. It starts with the first kick on the drop's bar line: a roll into the drop doesn't duck. TAP and STRIP set to ROLE KICK, PERC or DECK don't dip, like SF KICK, PERC, STRIKE and DECK.

LOOPS, RESTARTS AND JUMPS

Everything follows Live's song position. When it jumps (arrangement loop brace, stop and restart, clicking to a new position), kick reports from before the jump are ignored, so nothing ducks for a kick that is no longer playing. RUMBLE re-arms. Playing DECK tracks keep running and are re-anchored by whole beats, so they stay on the beat.

4 ALLOC: THE MIX ASSIGNER

ALLOC keeps sources from piling up in the same frequencies. You decide the order of importance; each frequency band has a priority curve over that order; the display shows what every source keeps and loses. Nothing is decided behind your back.

THE ORDER

— Every SF instrument, STRIP, TAP and DECK reports its energy in ten fixed bands and appears in the SOURCES list (by its Live track name with VST3, otherwise by role: PERC, PERC 2 ...).

— Drag the list into order, top = most important: the order for ALL BANDS. It is saved with the set and only changes when you drag it. New sources are added at the bottom.

— Low bands and high bands usually want different orders (kick and bass matter at 60 Hz, hats and voice at 3 kHz). Click a band to show its order in the list and drag: the band now has its OWN order. LINK TO ALL returns it to the global order.

— Each source applies its own cuts: audio never leaves its track.

THE CURVE (PER BAND)

— **PRIORITY** moves the peak of the curve up and down the list. The source at the peak is protected.

— **DEPTH** sets how hard everything away from the peak is cut, up to MAX CUT. At 0 the band is untouched.

— **Q** sets the width. Narrow: one source owns the band. Wide: the sources next to the peak share it.

— Each knob always does the same thing to the same band, so once set it stays predictable.

DYNAMIC OR STATIC

DYNAMIC (default): a source is cut only while a more favoured source is actually playing in that band. When the bass stops, the pad gets 180 Hz back within a fraction of a second. **STATIC**: the cuts stay on all the time, like a fixed EQ.

READING THE DISPLAY

— Each band is a column; each source a lane, in that band's order and named in the lane. A selected band has a white frame; OWN marks a band with its own order.

— **Grey**: what the source plays in that band after ALLOC. **Amber**: what ALLOC takes away.

— The white line is the curve (right = protected); it turns cyan while you turn one of that band's knobs.

— The name at the bottom of the band is the source the peak sits on.

THE BANDS

BAND	RANGE
1	20-60 Hz
2	60-100 Hz
3	100-150 Hz
4	150-220 Hz
5	220-320 Hz
6	320-480 Hz
7	480-750 Hz
8	750-1500 Hz
9	1.5-4 kHz
10	4-20 kHz

5 SOUND AND LEVELS

The sets are the releases, so every plugin is built to release quality: nothing you hear should come from the implementation. These figures are measured by the offline tests on every build.

GAIN STAGING

Every model of every instrument is calibrated to its role's loudness at default settings, so switching models never jumps the level and a rig with every LEVEL at its default is already a balanced rough mix with headroom into GLUE and LIMIT. SAMPLE models play the file at its own level.

ROLE	TARGET (LUFS, INTEGRATED)	ROLE	TARGET
KICK	-15	LEAD	-20
PERC	-22	PAD	-22
RUMBLE	-20	VOICE	-20
BASS	-18	STRIKE	-20

NO ALIASING

Every stage that distorts runs oversampled, all the time (switching an oversampler in and out would click), and every drive knob fades in from exactly clean: synth voices at 2x (the ladder, MS20-TYPE and SYNC models at 4x), KICK and PERC at 2x, DRIVE and STRIP at 4x, the drive in FILTER, ENERGY and VOICE, the saturation inside the DELAY loop, RUMBLE's GRIT, LOWEND's SUB CLIP, STRIKE's fracture and TONE at 2x, LIMIT's clipper at 4x. The filters reject images and aliases by 112 dB or more and are flat to 20 kHz. Oscillators are band-limited (wavetables per octave, corrected hard sync): measured alias products sit 77 to 105 dB below the note.

THE MASTER

— SF LIMIT measures true peak (the level between samples that a DAC or a codec will produce) at 8x, and its output never exceeds CEILING. Use -1 dB for venues and releases.

— Its meter follows ITU-R BS.1770-4 / EBU R128 exactly: MOMENTARY (400 ms), SHORT-TERM (3 s), INTEGRATED (gated, over the whole set since RESET). It passes the EBU Tech 3341 level, gating and short-term cases within 0.02 LU.

— SF LIMIT AUTO keeps the loudest moments of the last minute at TARGET and holds through breakdowns, so the drop comes back where it was. AUTO MODE DOWN ONLY (the default) never lifts the set above unity.

— SF TONE voices the whole set (LOW, MUD, PRESENCE, AIR, TILT); SF ROOM EQ corrects the room, per venue. Keep them separate: TONE is the record, ROOM EQ is the venue, and only the venue hears ROOM EQ (below).

ANY BUFFER SIZE, ANY SAMPLE RATE

Every plugin processes in 64-sample steps on a fixed grid of the audio stream, whatever Live's buffer size, so a set sounds exactly the same at buffer 64 and buffer 2048 (bit for bit) and at 44.1, 48 or 96 kHz. Song recalls land on the exact sample of the bar line.

RECORDING A SET

— Record the Main output (after SF LIMIT, before SF ROOM EQ) at the session's sample rate, 24-bit or 32-bit float: an audio track REC with Audio From = Resampling, Monitor Off, armed. Live records what comes into a track, before its devices.

— The room's correction stays out of the release: SF ROOM EQ sits on the PA track (see Setting up Live), or do the venue EQ in the audio interface's own mixer, after Live.

— Press RESET on SF LIMIT before you start: INTEGRATED then reads the loudness of the whole set.

— MOTION's RANDOM and DRIFT are seeded and the seeds are saved with the set, so they move the same way every time the set is played.

6 SETTING UP LIVE

TRACKS

WHERE	WHAT
MIDI track	SF CONTROL, input from the OXI One. Makes no sound.
MIDI tracks	SF KICK, SF PERC (2-3), SF RUMBLE (no clips needed), SF BASS, SF LEAD (1-2), SF VOICE, SF PAD, SF STRIKE, SF DECK.
Groups	SF ENERGY on each group, PROFILE to match (DRUMS, BASS, MUSIC, VOICE).
Returns	SF DELAY, SF REVERB, SF DENSITY, each with DRY off (the return gives only the effect).
Main (master)	SF ISOLATOR > SF BUILD > SF GLUE > SF TONE > SF LOWEND > SF LIMIT. SF ALLOC anywhere on it. Its output to an output pair you don't use.
Audio track PA	Audio From = Resampling (the Main output), Monitor In, SF ROOM EQ, Audio To = the PA outputs. The venue hears ROOM EQ.
Audio track REC	Audio From = Resampling, Monitor Off, armed: records the set without ROOM EQ.
Live instruments / hardware	SF STRIP last on the track: the suite's performance row, Pocket and ALLOC for any sound.
Stems	SF TAP with ROLE set (no sound of its own), or SF STRIP to play them.

THE E16

— Map with Live's MIDI Map (Cmd-M): unfold the plugin's parameter panel (the triangle in its title bar), click the parameter there and turn the E16 knob. A parameter that isn't listed: Configure, touch it in the plugin window, Configure off.

— In Live's MIDI preferences turn **Remote** on for the E16's Port 2, input and output. Live then sends values back, so the rings follow song recalls and ALLOC slot changes and the knobs hold the new values.

— Absolute 0-127 CC is right; the plugins smooth every step. Chapter 7 has a ready-made E16 scene for the show template, SF SHOW, and the show set with every address already mapped.

SAMPLES

Every instrument has a SAMPLE model. Drop an audio file on the plugin window or click LOAD. The sample is analysed (root note, decay, onsets) and embedded in the Live set, so the set travels without the file.

FIRST SESSION CHECKS

— SF CONTROL shows **BUS SHARED**. Play SF KICK: it shows **KICK REPORTING**.

— SF KICK + SF BASS with POCKET DEPTH up and SHAPE right: the pump lands exactly on the kick.

— SF ALLOC on the master: every SF source appears in the list; raise DEPTH on a band where two of them overlap and watch the amber.

— Store two songs; switch with Program Change: the E16 rings follow. With SF FILTER on tracks above and below SF CONTROL, every track changes on the same bar.

— SF DECK: load a track, PLAY A: it starts on the bar, locked to Live's tempo.

— Set KEY on SF CONTROL: BASS, LEAD and PAD transpose, KICK and RUMBLE retune. Hold PANIC with SF DELAY in feedback: silence within 5 ms, for as long as you hold it.

— Record a minute on REC with a deep cut on ROOM EQ: the PA has the cut, the recording doesn't.

Chapter 7 is a complete show set built on this layout.

7 THE SHOW TEMPLATE

One Live set for every show, built on fixed roles. The OXI ONE plays the notes and patterns; the E16 plays the sound, the balance and the transitions. The E16's top half is the same on every page, so the whole arrangement is always under one hand while the other works deep in one sound.

The set is built: **controllers/live/SF SHOW.als** in the repository holds every track, group and return below, every SF plugin with its starting settings, the routing, and every E16 address mapped. Open it, do the five steps its README lists (MIDI From, the outputs, the two keys, two options, the save), and keep it as your template.

TRACKS

TRACK	DEVICES	MIDI FROM
00 CONTROL	SF CONTROL	ONE port B, ch 16
01 KICK	SF KICK	ONE port A, ch 1
10 LOW (group)	SF ENERGY, PROFILE BASS	
11 BASS	SF BASS	port B, ch 1
12 RUMBLE	SF RUMBLE, SOURCE KICK BUS	none: it follows the kick
20 DRUMS (group)	SF ENERGY, PROFILE DRUMS; SF REPEAT	
21 CLOSED	SF PERC: closed hat	port A, ch 2
22 OPEN	SF PERC: open hat or ride	port A, ch 3
23 CLAP	SF PERC: clap or rim	port A, ch 4
24 STRIKE	SF STRIKE	port B, ch 6
30 MUSIC (group)	SF ENERGY, PROFILE MUSIC	
31 ACID	SF LEAD: ACID 303-TYPE or another mono lead	port B, ch 2
32 STAB	SF LEAD: chords and stabs	port B, ch 3
33 TEXTURE	SF PAD, often GRANULAR	port B, ch 4
34 VOICE	SF VOICE: phrases, radio fragments, choir	port B, ch 5
40 BRIDGE	Live audio clips	
41 DECK	SF DECK	port B, ch 7 (hot cues; optional)
A DUB (return)	SF DELAY, DRY off	
B SPACE (return)	SF REVERB, DRY off	
90 LIVE BUS	SF ISOLATOR > SF BUILD > SF GLUE > SF TONE > SF LOWEND; SF ALLOC (off at first)	
Main	SF LIMIT	
92 PA	SF ROOM EQ	audio: Resampling
93 REC	nothing	audio: Resampling

— The kick stays outside LOW, so bass and rumble fade together while the kick holds the floor. In each section decide whether BASS or RUMBLE carries the sustained low end: one of two knobs then steers it.

— The PERCs are separate instruments: CHOKE works inside one track only, so a closed hat doesn't cut the open hat on its own track. Write the open hat's rhythm and DECAY for that.

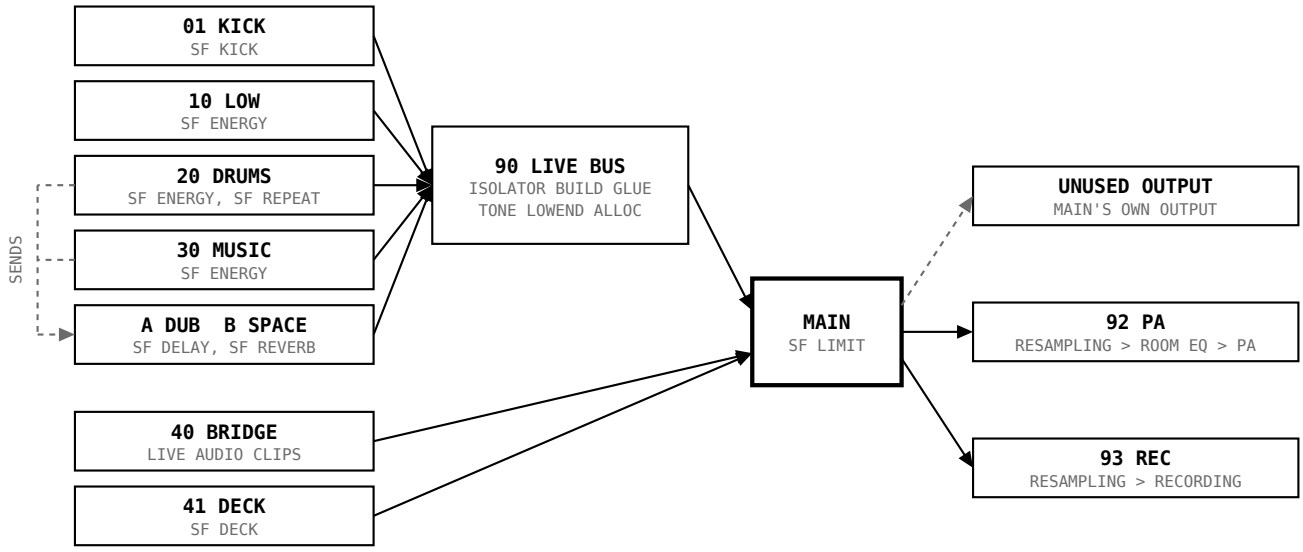
— STAB: if the ONE sends whole chords, set CHORD to NONE; if it sends single notes, CHORD (or the model's own chord) voices them. Not both.

— ACID 303-TYPE slides when notes overlap, always over 60 ms like a 303: write legato notes for slides. GLIDE doesn't change it.

— VOICE with a phrase loaded: C3 plays slice 1, C#3 slice 2 and so on; below C3, the whole phrase. Sequence it as phrase choreography, not as a pitched sampler.

— Name the tracks as above: SF ALLOC lists its sources by Live track name (VST3).

ROUTING



TRACK	INPUT / MONITOR	OUTPUT
Instrument tracks	MIDI From as above, Monitor In	their group; KICK to 90 LIVE BUS
Groups, returns A and B		90 LIVE BUS
90 LIVE BUS	the tracks routed to it, Monitor In, every send disabled	Main
40 BRIDGE	clips, Monitor Off (clips still play)	Main
41 DECK	MIDI From port B ch 7, Monitor In	Main
Main		an output pair that goes nowhere
92 PA	Audio From Resampling, Monitor In, sends disabled	the PA outputs
93 REC	Audio From Resampling, Monitor Off, sends disabled	Sends Only
Cue	Live's Cue Out; DECK's CUE track	the headphone outputs

— Main is the set. Exporting or resampling it gives the release mix; the venue's correction (ROOM EQ) exists only on 92 PA, so the recording never hears it.

— **Disable every send on 90 LIVE BUS** (right-click a send, Disable Send). The returns feed LIVE BUS, and Live can't compensate a return's delay on a track whose matching send is active, even at zero (Ableton's Delay Compensation FAQ).

— Pull 90 LIVE BUS down and the live instruments and their effects go while BRIDGE and DECK keep playing. Mute 92 PA and the speakers go silent while REC keeps recording. Put 92 PA's on/off on a computer key (Cmd-K, click its activator, press the key), not on an E16 press: a knob is easily pressed while it is turned.

— PANIC silences every SF plugin, DECK included, but not Live's own clips and devices: for total silence, mute 92 PA.

— BRIDGE is for transitions and regrouping. It still needs Live: it is not a backup for a computer failure.

— ROOM EQ only cuts, so it needs no limiter after it; LIMIT's -1 dB ceiling leaves the margin.

THE MASTER DEVICES

— Start GLUE, TONE and LOWEND gently: the set doesn't need every processor working all the time. Soundcheck the busiest passage of the night and set LIMIT's INPUT for 1-3 dB of limiting there.

— LIMIT AUTO off while you set the gain staging, then on, AUTO MODE DOWN ONLY, for shows: it stops the set creeping above TARGET and holds through breakdowns.

— SF ALLOC off at first. Turn it on band by band once the arrangement is known. It measures each source before its LEVEL and before Live's fader, so a source you've faded out still claims its bands: take a source out with its lane mute or its FILTER, or switch its ALLOC off.

THE OXI ONE

Port A carries the drums, port B everything pitched and SF CONTROL. Every sequencer has its own channel; in Live, set every track's MIDI From to its port and channel, never All Ins / All Channels.

SEQUENCER	MODE	OUTPUT	TRACK
1 DRUMS	Multitrack: lane 1-4 on ch 1-4	port A	KICK, CLOSED, OPEN, CLAP (lanes 5-8 spare)
2 BASS	Mono	port B ch 1	BASS
3 ACID	Mono	port B ch 2	ACID
4 STAB	Chord or Poly	port B ch 3	STAB
5 TEXTURE	Poly	port B ch 4	TEXTURE
6 VOICE	Mono	port B ch 5	VOICE
7 STRIKE	Mono to start	port B ch 6	STRIKE
8 CONTROL	Program Change only	port B ch 16	CONTROL

— KICK lane on C1: with KEY TRACK on, any other note transposes the kick (C1 = TUNE), so C1 is right either way. PERC plays any note.

— RUMBLE needs no lane: SOURCE KICK BUS follows SF KICK.

— KEY OFF on SF CONTROL to start with: the ONE's notes set the pitch, and KICK and RUMBLE are tuned per song with TUNE. Turn KEY on for songs that should change key without reprogramming the ONE (mind the low register, chapter 1).

— One ONE project per track, with a pattern per section: SPARSE, MAIN, VARIATION, EXIT. Sequencer 8 sends that track's Program Change on port B ch 16, so loading the project recalls the matching SF song. Program Change stays off on the musical sequencers.

— When patterns change is set on the ONE (its launch and arranger settings): Live's launch quantization doesn't reach it.

CLOCK, MIDI AND LATENCY

LIVE MIDI PORT	TRACK	SYNC	REMOTE
ONE port A in	on	off	off
ONE port B in	on	off	off
ONE out (the one it takes clock from)	off	on	off
E16 Port 2 in	off	off	on
E16 Port 2 out	off	off	on
E16 Ports 1 and 3, in and out	off	off	off

— Live is the clock master. On the ONE: USB device mode, sync from USB, Auto Sync on (it starts with Live), Start and Stop accepted, MIDI Thru off. Never send clock both ways.

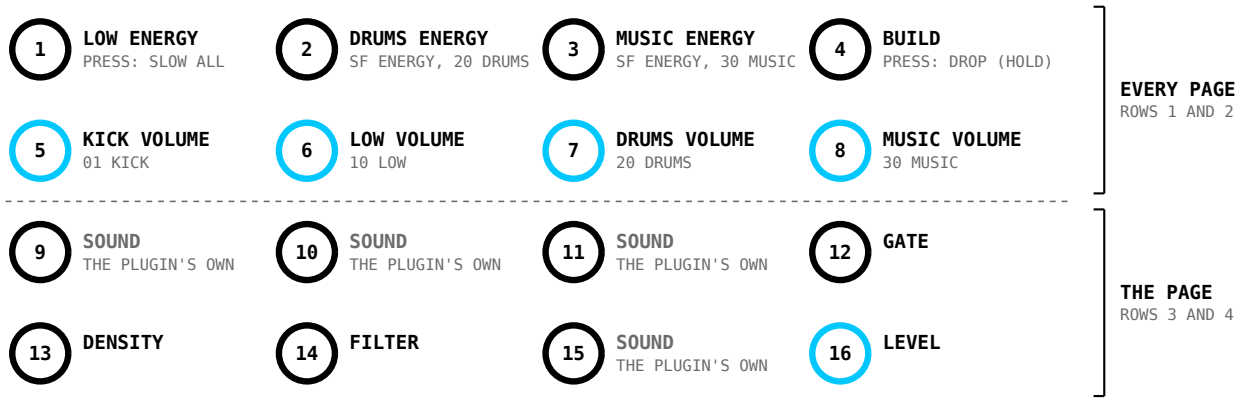
— Audio: 48 kHz, buffer 256 (Pocket needs 512 or less), Delay Compensation on, **Reduced Latency When Monitoring off** (it skips compensation on monitored tracks, so returns and Pocket drift), launch quantization 1 bar, record quantization off.

— Calibrate once: record a plain quarter-note pattern from the ONE, compare it with Live's grid and set the ONE port's MIDI Clock Sync Delay in Live's preferences. Leave POCKET OFFSET at 0 unless the pump is still off after that.

— Every Pocket stage in a chain adds its 512-sample lookahead to the total latency (compensated, but it adds up): no SF TAP or SF DUCK on tracks that already hold SF instruments.

THE E16

One E16 scene runs the whole rig: **SF SHOW**, in the repository under controllers/e16, with its address map (sf-show-map.csv) and loading steps (README); SF SHOW.als has every address mapped. The E16 is a 4 × 4 grid, numbered row by row from the top left. Rows 1 and 2 are the same on every page; rows 3 and 4 are the page.



ROWS 3 AND 4 AS ON AN INSTRUMENT PAGE. LIGHT BLUE RING: A VOLUME.

— **Press 1:** SLOW ALL (swap): a quick press flips the knob to SLOW ALL on SF CONTROL (green ring): how long your next moves take, OFF to 64 BAR; another press flips it back.

— **Press 4:** DROP (hold): the drop lands on the next bar line (BUILD's GAP before it).

— **PANIC and 92 PA's on/off are on computer keys** (Key Map, Cmd-K), never on a press: a knob is easily pressed while it is turned, and neither may ever happen by accident. A key flips them: PANIC on, every SF plugin silent; again, back. Every press on the E16 is one the set plays through if it happens by accident.

— The ENERGY knobs are the group DJ filters, plus drive, space, echo and chop: centre is neutral, left takes energy out, right puts it in.

— Each instrument's LEVEL belongs to its song; the volumes here are your balance on the night. Set every LEVEL at soundcheck so a volume at 0 dB gives the intended mix, and give every volume mapping a maximum of 0 dB (Max in Live's MIDI Mappings).

— Rings: light blue is a volume, white is everything else, green is knob 1 on SLOW ALL. A knob that is travelling (SLOW) turns green and walks with the sound until it arrives (GLIDE RINGS: SF CONTROL's E16 RINGS and the scene's script; controllers/e16 README).

THE E16: PAGES

PAGE	ROW 3: ENCODERS 9-12	ROW 4: ENCODERS 13-16	PRESSES
1 PERFORM	ACID CUTOFF / STAB SEND A / TEXTURE COLOR / VOICE FILTER	DUB RETURN / SPACE RETURN / LIVE BUS VOLUME / BRIDGE VOLUME	
2 KICK	TUNE / DECAY / PUNCH / CLICK	DENSITY / FILTER / DRIVE / LEVEL	
3 BASS	CUTOFF / RES / ENV AMT / GATE	DENSITY / FILTER / DRIVE / LEVEL	
4 RUMBLE	SIZE / DARK / GRIT / GATE	DENSITY / FILTER / PUMP / LEVEL	
5 HATS	CLOSED DENSITY / CLOSED DECAY / CLOSED TONE / OPEN DENSITY	OPEN DECAY / OPEN TONE / CLAP DENSITY / CLAP DECAY	
6 STRIKE	POWER / DISTANCE / SIZE / DECAY	DENSITY / FILTER / MATERIAL / LEVEL	
7 ACID	CUTOFF / RES / ENV AMT / GATE	DENSITY / FILTER / DRIVE / LEVEL	
8 STAB	CUTOFF / RES / ENV AMT / GATE	DENSITY / FILTER / SPACE / LEVEL	
9 TEXTURE	CUTOFF / COLOR / DETUNE / GATE	DENSITY / FILTER / ATTACK / LEVEL	
10 VOICE	SIGNAL / TONE / ECHO / GATE	DENSITY / FILTER / DRIVE / LEVEL	
11 SPACE	DUB TIME / DUB FEEDBACK / DUB TONE / DUB DRIVE	SPACE SIZE / SPACE DECAY / SPACE DARK / SPACE DUCK	9 THROW (hold), 10 FREEZE (toggle), 13 THROW (hold), 14 FREEZE (toggle)
12 MACROS	MACRO 1 / MACRO 2 / MACRO 3 / MACRO 4	MACRO 5 / MACRO 6 / REPEAT LENGTH / REPEAT PITCH	15 ROLL (hold)

— On every single-instrument page 13 is DENSITY, 14 FILTER and 16 LEVEL, and 12 is GATE wherever there is one: the hand learns one layout (HATS holds three PERCs). The rest is the plugin's own performance row, with a swap where a model's character lives elsewhere (GRANULAR: COLOR is grain size, DETUNE is scatter; STAB: SPACE is the ensemble).

— On SPACE the presses sit under their knobs: row 3 is DUB, row 4 is SPACE. On MACROS, hold ROLL on 15 and turn LENGTH under the same finger.

— The HATS page is the fill page: CLOSED DENSITY with ECHO SPACE 1/16 turns 1/8 hats into 1/16s, the send-delay trick built in.

THE E16: ADDRESSES AND SETTINGS

— **Channel = track:** 1 KICK, 2 BASS, 3 RUMBLE, 4 CLOSED, 5 OPEN, 6 CLAP, 7 STRIKE, 8 ACID, 9 STAB, 10 TEXTURE, 11 VOICE, 12 DUB, 13 SPACE, 14 the groups (LOW, DRUMS, MUSIC and DRUMS' REPEAT), 15 LIVE BUS and BRIDGE, 16 CONTROL.

— **CC = block base + the parameter's place in its plugin's own order**, counting from 0. One plugin on a channel: base 20, so its performance row (its first eight parameters) is CC 20-27. Channel 14: LOW ENERGY 20, DRUMS ENERGY 30, MUSIC ENERGY 40, REPEAT 50. Channel 15: BUILD 20

(DUCK TIME, its eleventh, has no address), with 30-79 kept for the master devices. Channel 16: CONTROL 20 (MACRO 1-6 are 20-25). The plugins never reorder their parameters, so an address never moves. The appendix lists every address.

— **Mixer:** CC 100 volume, 101 send A, 102 send B. Channel 14: 100-102 the LOW, DRUMS and MUSIC volumes; channel 15: 100 LIVE BUS, 101 BRIDGE (102 kept for DECK). A parameter on two pages sends the same address from both: two ways to one mapping.

— **Presses:** a hold is a CC, 127 on press and 0 on release, so the parameter is on while the knob is down. A toggle is a note: Live flips a switch on every Note On, so the E16 can never think FREEZE is on while Live has it off. SF SHOW sets every turn; until the OXI App's press format is known, the presses are added in the App (the README lists them), and again after loading a new scene file.

— **On the E16:** Auto Transmit off, for the scene and globally (stored values sent on a page change would pull Live's faders to them), MIDI Thru off. Stay in SF SHOW for the whole show.

— **In Live:** E16 Port 2 Remote on, input and output; Ports 1 and 3 off (Port 1 is the OXI App's, and a second Remote port doubles every message: a toggle would flip twice). Takeover Mode: Value Scaling, so a knob that disagrees with Live after a recall moves the sound smoothly instead of jumping.

— **Mapped once, kept:** MIDI mappings belong to the Live set. SF SHOW.als comes with every address mapped, every volume at Max 0 dB and DUB FEEDBACK at 0-70 %; save it as a template (File > Save Live Set As Template) and new sets start mapped. In a set of your own, map once from the appendix and do the same.

— **Map the E16 straight to the SF parameters, not through Instrument Rack macros.** A song recall changes the plugin's value underneath a macro: the macro doesn't follow, its ring shows the wrong value and the next turn jumps the sound back. Ranges go in Live's MIDI Mappings (Min and Max per mapping): give resonance and feedback the part of their travel that matters, and widen it when the music asks.

— Check the shared knobs once: turn LOW ENERGY on page 2, go to page 1 and turn it a step. The sound must move a step, not jump back. If it jumps, the README's relative fallback fixes it.

— Keep STORE ALL, MODEL, KEY and the calibration controls off the performance pages.

MACROS

SF CONTROL's six macros, on the MACROS page. Every SF plugin follows one macro (MACRO SRC) by an amount (MACRO AMT, negative turns it the other way); a macro moves the knobs marked M in the module tables, adds to them and is stored per song. A starting plan:

MACRO	FOLLOWED BY	MOVES
1 ENERGY	the three ENERGY plugins, BUILD	every group's energy at once (a negative amount on LOW darkens the bass while the rest lifts)
2 MOOD	TEXTURE, STAB, VOICE	cutoff, envelope and drive; the voice's SIGNAL and ECHO
3 DENSITY	CLOSED, OPEN, CLAP, STRIKE	DECAY and DENSITY: the fills
4 PULSE	KICK, RUMBLE	kick DECAY, PUNCH, DRIVE; rumble SIZE, GRIT, DENSITY
5 SPACE	DUB, SPACE returns	delay FEEDBACK and MIX; reverb SIZE, DECAY and MIX
6 FRACTURE	ACID, BASS	cutoff, envelope, density and drive: smooth to heavy

Start the amounts at 30-50 %. One macro per plugin: a plugin can follow only one.

RETURNS AND POCKET

— A DUB: SF DELAY, DRY off, TIME synced, FEEDBACK mapped to 0-70 % at first, DUCK up so the repeats sit around the kick. INPUT THROW ONLY makes its THROW press the only way in: dub throws.

— B SPACE: SF REVERB, DRY off, DUCK 30-60 % so the tail blooms in the gaps, LO CUT up.

— Sends post-fader: turning a source down stops feeding its effect while the tail rings on. No return-to-return sends at first.

— Pocket needs no wiring: SF KICK reports on the bus. SF CONTROL should show BUS SHARED and KICK REPORTING.

— Pulling the KICK fader down leaves the kick playing on the bus: Pocket and RUMBLE keep pumping to a silent kick, which can be the point. For a clean withdrawal, mute the kick lane on the ONE.

SONGS, PATTERNS AND SCENES

SYSTEM	RESPONSIBLE FOR
OXI ONE projects and patterns	Notes, rhythms, gates, accents, the pattern progression.
SF songs (SF CONTROL)	The sound of every SF plugin, recalled on the bar.
E16 pages	Where your hands are.
Live Session scenes	BRIDGE clips, reference material, rehearsal.
Live Arrangement	Recording the performance.

— The roles stay put for the whole set: a new track changes what they play, not where they are.

— Rehearse song recall, SF REPEAT and SF BUILD's drop (GAP, TAIL, DROP DUCK) in your Live before relying on them in a transition: all three changed on Nov 3.

RECORDING AND REHEARSAL

— Instrument tracks on Monitor In: they play from the ONE whether or not they're armed.

— To record: RESET on SF LIMIT, arm the instrument tracks and 93 REC, Automation Arm on, Arrangement record, play. The stereo recording on REC is the set; the MIDI and automation are for editing and rehearsal.

— To replay recorded MIDI: stop the ONE, set those tracks to Monitor Auto and disarm them (Monitor In ignores a track's own clips).

— Collect All and Save. SF samples are inside the set; DECK tracks are referenced by path, so keep the files where they are.

BUILD ORDER

1. SF SHOW.als (controllers/live): MIDI From, the outputs, PANIC and 92 PA on their keys, the two options; saved as a template.
2. CONTROL, KICK, BASS, RUMBLE, LIVE BUS, PA and REC playing. Clock and latency calibrated, a short recording made.
3. CLOSED, OPEN, CLAP and STRIKE.
4. ACID, STAB, TEXTURE and VOICE, one at a time.
5. The returns; check the sends are disabled on LIVE BUS.
6. BRIDGE and DECK.
7. SF SHOW on the E16 (controllers/e16): the scene loaded, the presses added, every page checked against the set's mappings; the macros.
8. Rehearse moving between BRIDGE, DECK and LIVE BUS.
9. Record a whole set before adding anything else.

BEFORE THE FIRST SHOW

— CONTROL shows BUS SHARED and KICK REPORTING; the pump lands on the kick.

— The ONE's Program Change recalls the song, every track on the same bar, and a recalled knob is yours the moment you turn it. (VST3 needs Live 12.2 or later; the AU works in any version.)

— PANIC from its key: silence, and every header shows PANIC. Again: back without a click. 92 PA muted from its key: the speakers silent, REC still recording.

— Every E16 hold press (DROP, THROW, ROLL) held for three seconds stays on for the whole hold and goes off on release.

— The shared E16 knobs don't jump: LOW ENERGY turned on page 2, then a step on page 1, moves a step. Changing pages sends nothing (Live's MIDI input indicator stays dark).

— A minute recorded on REC with a deep ROOM EQ cut: the cut isn't in the recording.

8 INSTRUMENTS

SF KICK

INSTRUMENT, MIDI TRACK

Kick drum with six synthesised models and a sampler. Every hit is reported to the other SF plugins, which duck around it (Pocket).



USING IT

— Put it on a MIDI track. Any note plays the kick (KEY TRACK off).

— DECAY is your GATE: short and tight for driving sections, long and boomy for breakdowns. RUMBLE and the bass pockets follow the change automatically.

— Ride DENSITY with ECHO SPACE at 1/16 or 1/8D to thicken the kick into a drop. Ghost kicks get smaller pockets.

— High-passing the kick with FILTER hands the sub to the bass and rumble: their pockets close as the kick thins.

— Set the song's KEY on SF CONTROL and the kick tunes itself to the key root (KEY FOLLOW).

MODELS

NAME	SOUND
KICK 909-TYPE	Fast pitch drop, noise click, short body. The default techno kick.
KICK 808-TYPE	Low, long, tuned boom with little click.
KICK SINE	Clean sine with a moderate sweep. Deep and minimal (dub techno).
KICK INDUSTRIAL	Steep sweep, heavy drive and clip. Hard and compressed.
KICK MODULAR	Triangle core with noise FM and a resonant low-pass sweep (DFAM-like).
KICK DUB	Soft, low, slow sweep, filtered dark.
SAMPLE	Plays the dropped one-shot. Its pitch and decay are analysed and reported for Pocket.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
TUNE	-12.0 st to 12.0 st default 0.0 st	Kick pitch in semitones around the model's fundamental (about 44-55 Hz).
DECAY M	0 % to 100 % default 50 %	Body length, x0.25 to x4 of the model's decay. This is the kick's GATE. On SAMPLE it gates the sample: 0 = 10 ms, 90 % and up = the whole sample.
PUNCH M	0 % to 100 % default 50 %	Depth of the pitch sweep. More = harder transient, less = rounder.
CLICK	0 % to 100 % default 50 %	Level of the noise/impulse click on the attack.
DRIVE M	0 % to 100 % default 20 %	Saturation on the kick, added to the model's own.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
DENSITY	0 % to 100 % default 0 %	Ghost notes. 0-50 %: one echo of every note fades in one ECHO SPACE later (1/8 hats become 1/16 hats with ECHO SPACE at 1/16). 50-100 %: two more ghosts at half that spacing (1/32 rolls). Each ghost is quieter, darker and shorter than the note before it. Ghosts are played softer and also faded in after the instrument's velocity response, so at the first notch they are 30 dB down: a fill really rises from silence. A quiet ghost never cuts the ringing kick or chokes the previous hit.
LEVEL	-36.0 dB to 6.0 dB default -3.0 dB	Output level. Default is set so the module sits sensibly in a mix.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
LAYER MODEL	KICK 909-TYPE, KICK 808-TYPE, KICK SINE, KICK INDUSTRIAL, KICK MODULAR, KICK DUB, SAMPLE default KICK 808-TYPE	Second model hit with every note (e.g. 909 click on an 808 body).
LAYER	0 % to 100 % default 0 %	Level of the LAYER MODEL. Main and layer always add up to the level of one kick.
PITCH TIME	0 % to 100 % default 50 %	How fast the pitch sweep falls. Short = snappy, long = laser.
VEL SENS	0 % to 100 % default 50 %	How much velocity changes the level. 0 = every hit equal.
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO SWING	0 % to 100 % default 0 %	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
KEY TRACK	ON / OFF default OFF	On: the note number sets the pitch (C1 = TUNE). Off: every note plays the same kick.
FILTER RES	0 % to 100 % default 10 %	Resonance of the output FILTER.
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: with a KEY set on SF CONTROL, the model's fundamental lands on the nearest key root and TUNE moves from there (TUNE 0 = on the root). Off: ignores KEY.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF PERC

INSTRUMENT, MIDI TRACK

Hats, cymbals, claps and percussion. Thirteen models including a sampler. Run two or three instances for a full kit.



USING IT

— Program fixed 1/8 hats on the OXI. Bring DENSITY up with ECHO SPACE 1/16 to turn them into 1/16s without changing the pattern; above 50 % they start rolling.

— DECAY opens closed hats into open hats.

— ECHO PROB under 100 % makes the fills irregular; SPREAD throws the ghosts left and right.

— For the real 909 hat sound, load your own samples into the SAMPLE model.

MODELS

NAME	SOUND
HAT 909-TYPE	Six square oscillators plus noise, bright band-pass. Synthesised approximation of the sampled original.
HAT 808-TYPE	Six-oscillator metallic hat, darker, less noise.
HAT 606-TYPE	Thin, noisy, ticky.
RIDE 909-TYPE	Long metallic ride.
CRASH	Noise-heavy long crash.
CLAP	Three quick noise bursts and a tail.
RIM	Two tuned sines, short.
COWBELL	Two square oscillators through a band-pass, two-stage decay.
CLAVE	Short tuned sine.
SHAKER	Band-passed noise with a soft attack.
METAL FM	Inharmonic two-operator FM hit. Industrial.
MODAL	Six tuned resonators struck by a noise burst. Wood, glass or metal depending on TONE.
SAMPLE	Plays the dropped one-shot, pitched by TUNE.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
TUNE	-24.0 st to 24.0 st default 0.0 st	Pitch in semitones (-24 to +24).
DECAY M	0 % to 100 % default 50 %	Length of each hit, x0.25 to x4. Closed to open. On SAMPLE it gates the sample: 0 = 10 ms, 90 % and up = the whole sample.
TONE	0 % to 100 % default 50 %	Model-specific brightness: band-pass position on hats, oscillator spread on FM, resonator ratios on MODAL.
DENSITY M	0 % to 100 % default 0 %	Ghost notes. 0-50 %: one echo of every note fades in one ECHO SPACE later (1/8 hats become 1/16 hats with ECHO SPACE at 1/16). 50-100 %: two more ghosts at half that spacing (1/32 rolls). Each ghost is quieter, darker and shorter than the note before it. Ghosts are played softer and also faded in after the instrument's velocity response, so at the first notch they are 30 dB down: a fill really rises from silence. A quiet ghost never cuts the ringing kick or chokes the previous hit.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
ACCENT	0 % to 100 % default 50 %	How much velocity matters. 0 = all hits equal.
DRIVE	0 % to 100 % default 0 %	Saturation on each hit.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO SWING	0 % to 100 % default 0 %	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
ECHO PROB	0 % to 100 % default 100 %	Chance each ghost note plays. Below 100 % the fills become irregular, like a drummer.
CHOKE	ON / OFF default ON	On: a new hit cuts the previous one (like a closed hat choking an open hat).
SPREAD	0 % to 100 % default 30 %	Pans the DENSITY ghosts alternately left and right.
FILTER RES	0 % to 100 % default 10 %	Resonance of the output FILTER. Higher gives a sharper, more vocal sweep.
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF RUMBLE

INSTRUMENT, MIDI TRACK LATENCY 512 SAMPLES AT 48 KHZ

The kick-driven low-end bed of hypnotic techno. Needs no notes: it listens to SF KICK, excites a thump with the kick's own pitch sweep, and runs it through a reverb, delay or drone model, then darkens, dirties and ducks it.



USING IT

— Put it on a MIDI track (no clips needed) with SOURCE on KICK BUS.

— SIZE and DARK set the room; GRIT adds the dirt; PUMP decides how hard it ducks out of the kick's way.

— GATE below 100 % cuts the tail before the next kick: tight, rhythmic rumble.

— DENSITY adds extra thumps between the kicks for a rolling bed.

MODELS

NAME	SOUND
REVERB	Thump into a dark reverb. The classic.
DELAY	Thump into a 3/16 feedback delay.
SUB DRONE	Continuous sine at the kick's pitch, ducked by the kick: swells between hits.
DISTORTED	Reverb with heavy drive.
DUB BOOM	Delay and reverb together, softer.
SAMPLE	Plays a dropped rumble one-shot on every kick.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SIZE M	0 % to 100 % default 60 %	Reverb size / delay feedback.
DARK	0 % to 100 % default 60 %	Low-pass on the bed: 600 Hz down to 70 Hz.
GRIT M	0 % to 100 % default 30 %	Distortion on the bed.
PUMP	0 % to 100 % default 70 %	Pocket depth: how far the bed ducks under each kick.
GATE	0 % to 100 % default 100 %	Cuts the tail after each kick. 100 % = no gate; lower = shorter.
DENSITY M	0 % to 100 % default 0 %	Extra thumps between kicks (offbeat, then 16ths).
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
TUNE	-12.0 st to 12.0 st default 0.0 st	Pitch of the thump relative to the kick.
SOURCE	KICK BUS, MIDI, BOTH default KICK BUS	KICK BUS: triggered by SF KICK. MIDI: by notes on this track. BOTH.
PREDELAY	0, 1/32, 1/16, 1/8 default 1/32	Delay before the reverb: 0, 1/32, 1/16, 1/8.
WIDTH	0 % to 100 % default 30 %	Stereo width of the bed.
POCKET SHAPE	0 % to 100 % default 50 %	Pocket SHAPE (see Pocket chapter).
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/8	Spacing of DENSITY thumps.
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: with a KEY set on SF CONTROL, the thump lands on the nearest key root. Off: it follows the kick's pitch.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF BASS

INSTRUMENT, MIDI TRACK LATENCY 512 SAMPLES AT 48 KHZ

Mono bass synth: ladder, 101, sub-oscillator, acid, FM, sine and Berlin off-beat models. Shared engine with BASS / LEAD / PAD: the same controls on every model.



USING IT

— CUTOFF, RES and ENV AMT are the performance; one note can carry a whole track.

— POCKET DEPTH (default 60 %) keeps the bass out of the kick's way; it follows the kick's pitch and length.

— DENSITY ghost notes fill the gaps between notes but never interrupt a note that is held.

MODELS

NAME	SOUND
BASS LADDER	Two detuned saws and a sub through a Moog-style ladder.
MONO 101-TYPE	Saw + pulse + sub, 4-pole filter.
SUB OSC	Square plus heavy sub-oscillator, soft filter. Deep and round.
ACID 303-TYPE	Mono, resonant ladder, accent and slide (see LEAD).
FM	Two-operator FM bass; COLOR sets the index.
SINE SUB	Pure sine with slight drive. No filter.
OFFBEAT	Square + saw pluck, high resonance. The rubbery Berlin off-beat bass.
SAMPLE	Chromatic sampler from the analysed root note.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
CUTOFF M	0 % to 100 % default 35 %	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
RES	0 % to 100 % default 30 %	Resonance of the model's filter. On ACID 303-TYPE this is the squelch.
ENV AMT M	0 % to 100 % default 40 %	How far the filter envelope opens the filter on each note. Accented notes open further.
GATE	0 % to 100 % default 50 %	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
DENSITY M	0 % to 100 % default 0 %	Ghost notes. 0-50 %: one echo of every note fades in one ECHO SPACE later (1/8 hats become 1/16 hats with ECHO SPACE at 1/16). 50-100 %: two more ghosts at half that spacing (1/32 rolls). Each ghost is quieter, darker and shorter than the note before it. Ghosts are played softer and also faded in after the instrument's velocity response, so at the first notch they are 30 dB down: a fill really rises from silence. A quiet ghost never cuts the ringing kick or chokes the previous hit.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
DRIVE M	0 % to 100 % default 0 %	Saturation before the filter. Adds weight at low settings, grit at high.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
DECAY	0 % to 100 % default 50 %	Length of the filter envelope (and amp decay on most models). Short = plucky, long = sweeping. ACID: the classic decay knob.
ATTACK	0 % to 100 % default 0 %	Adds attack time to amp and filter envelopes. Up for swells.
ACCENT	0 % to 100 % default 50 %	How strongly accented notes (velocity above about 108) stand out: louder, brighter, shorter filter decay.
GLIDE	0 % to 100 % default 20 %	Portamento between notes (non-acid models). ACID slides only when notes overlap, like a 303.
TUNE	-24.0 st to 24.0 st default 0.0 st	Pitch offset in semitones (-24 to +24).
DETUNE	0 % to 100 % default 50 %	Spread between oscillators and unison voices. Also sets grain scatter on GRANULAR.
COLOR	0 % to 100 % default 50 %	Model-specific tone control. Mono subtractive: sub-oscillator level. ACID: below 50 % saw, above 50 % square. ORGAN: drawbar brightness. FM: modulation index. SYNC: sync sweep. WAVETABLE: table position (sine > saw > square > vocal). GRANULAR: grain size. NOISE BED: resonance of the noise.
LAYER MODEL	BASS LADDER, MONO 101-TYPE, SUB OSC, ACID 303-TYPE, FM, SINE SUB, OFFBEAT, SAMPLE default MONO 101-TYPE	Second model played with every note, blended by LAYER.
LAYER	0 % to 100 % default 0 %	Blend of the LAYER MODEL under the main model. Use it to put a sub under a stab or an acid line on a bass.
CHORD	MODEL, NONE, MIN7, MIN9, SUS4, MAJ7 ... (9) default MODEL	Notes played per key. MODEL uses the model's own chord (STAB DUB = minor 7, STAB DETROIT = minor 9, ORGAN = minor triad). NONE forces single notes. Or choose MIN7, MIN9, SUS4, MAJ7, MIN, 5TH, OCT.
SPACE	0 % to 100 % default 50 %	Ensemble chorus amount, for the models that have one (strings, poly pad, stabs, hoover, organ).
POCKET DEPTH	0 % to 100 % default 60 %	How deep this module ducks around SF KICK (Pocket). 0 = no ducking.
POCKET SHAPE	0 % to 100 % default 25 %	One knob from transparent to pumping. Left: a narrow cut that follows the kick's pitch sweep (felt more than heard). Middle: the low band ducks for the kick's length. Right: full-band pump.
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO SWING	0 % to 100 % default 0 %	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: follows the KEY set on SF CONTROL (parts written in C play in the song's key). Off: ignores KEY.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF LEAD

INSTRUMENT, MIDI TRACK LATENCY 512 SAMPLES AT 48 KHZ

Leads, acid lines and chord stabs. Shared engine with BASS / LEAD / PAD: the same controls on every model.



USING IT

— For acid: ACID 303-TYPE, RES high, ENV AMT and DECAY on knobs, accents and slides from the OXI.

— For dub stabs: STAB DUB with GATE short and SF DELAY on a return.

— CHORD overrides the model's chord; NONE gives single notes.

MODELS

NAME	SOUND
ACID 303-TYPE	Mono, resonant 4-pole ladder. Velocity above ~108 is an accent (louder, brighter, snappier). Overlapping notes slide without retriggering. COLOR: saw or square.
MONO 101-TYPE	Saw + pulse + sub. Classic mono sequence lead.
MS20-TYPE	Driven high-pass + low-pass pair, screaming resonance.
STAB DUB	Minor 7 chord stab, short filter envelope, light ensemble. Send it into SF DELAY.
STAB DETROIT	Minor 9 string/brass stab with unison and ensemble.
ORGAN M1-TYPE	Additive organ with key click and percussion, minor triad.
HOOVER	Detuned pulse-width-modulated stack with a pitch dive and heavy chorus.
FM BELL	Two-operator bell, long release.
SYNC	Hard-sync lead; COLOR and ENV AMT sweep the sync.
SAMPLE	Chromatic sampler from the analysed root note.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
CUTOFF M	0 % to 100 % default 40 %	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
RES	0 % to 100 % default 50 %	Resonance of the model's filter. On ACID 303-TYPE this is the squelch.
ENV AMT M	0 % to 100 % default 50 %	How far the filter envelope opens the filter on each note. Accented notes open further.
GATE	0 % to 100 % default 50 %	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
DENSITY M	0 % to 100 % default 0 %	Ghost notes. 0-50 %: one echo of every note fades in one ECHO SPACE later (1/8 hats become 1/16 hats with ECHO SPACE at 1/16). 50-100 %: two more ghosts at half that spacing (1/32 rolls). Each ghost is quieter, darker and shorter than the note before it. Ghosts are played softer and also faded in after the instrument's velocity response, so at the first notch they are 30 dB down: a fill really rises from silence. A quiet ghost never cuts the ringing kick or chokes the previous hit.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
DRIVE M	0 % to 100 % default 0 %	Saturation before the filter. Adds weight at low settings, grit at high.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
DECAY	0 % to 100 % default 50 %	Length of the filter envelope (and amp decay on most models). Short = plucky, long = sweeping. ACID: the classic decay knob.
ATTACK	0 % to 100 % default 0 %	Adds attack time to amp and filter envelopes. Up for swells.
ACCENT	0 % to 100 % default 50 %	How strongly accented notes (velocity above about 108) stand out: louder, brighter, shorter filter decay.
GLIDE	0 % to 100 % default 20 %	Portamento between notes (non-acid models). ACID slides only when notes overlap, like a 303.
TUNE	-24.0 st to 24.0 st default 0.0 st	Pitch offset in semitones (-24 to +24).
DETUNE	0 % to 100 % default 50 %	Spread between oscillators and unison voices. Also sets grain scatter on GRANULAR.
COLOR	0 % to 100 % default 50 %	Model-specific tone control. Mono subtractive: sub-oscillator level. ACID: below 50 % saw, above 50 % square. ORGAN: drawbar brightness. FM: modulation index. SYNC: sync sweep. WAVETABLE: table position (sine > saw > square > vocal). GRANULAR: grain size. NOISE BED: resonance of the noise.
LAYER MODEL	ACID 303-TYPE, MONO 101-TYPE, MS20-TYPE, STAB DUB, STAB DETROIT, ORGAN M1-TYPE ... (10) default MONO 101-TYPE	Second model played with every note, blended by LAYER.
LAYER	0 % to 100 % default 0 %	Blend of the LAYER MODEL under the main model. Use it to put a sub under a stab or an acid line on a bass.
CHORD	MODEL, NONE, MIN7, MIN9, SUS4, MAJ7 ... (9) default MODEL	Notes played per key. MODEL uses the model's own chord (STAB DUB = minor 7, STAB DETROIT = minor 9, ORGAN = minor triad). NONE forces single notes. Or choose MIN7, MIN9, SUS4, MAJ7, MIN, 5TH, OCT.
SPACE	0 % to 100 % default 50 %	Ensemble chorus amount, for the models that have one (strings, poly pad, stabs, hoover, organ).
POCKET DEPTH	0 % to 100 % default 20 %	How deep this module ducks around SF KICK (Pocket). 0 = no ducking.
POCKET SHAPE	0 % to 100 % default 25 %	One knob from transparent to pumping. Left: a narrow cut that follows the kick's pitch sweep (felt more than heard). Middle: the low band ducks for the kick's length. Right: full-band pump.
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO SWING	0 % to 100 % default 0 %	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: follows the KEY set on SF CONTROL (parts written in C play in the song's key). Off: ignores KEY.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF PAD

INSTRUMENT, MIDI TRACK LATENCY 512 SAMPLES AT 48 KHZ

Pads, drones and textures. Polyphonic, slow envelopes. Shared engine with BASS / LEAD / PAD: the same controls on every model.



USING IT

— GATE high keeps chords ringing; SLOW mode on CUTOFF gives long, even swells.

— GRANULAR and SAMPLE use a dropped file. Field recordings work well in GRANULAR.

— POCKET DEPTH lets the pad breathe with the kick without a sidechain.

MODELS

NAME	SOUND
STRING MACHINE	Four-voice unison saws plus an octave, deep ensemble.
POLY PAD	Saw + pulse + sub, slow filter envelope, chorus (Juno-like).
WAVETABLE	Morphs sine > saw > square > vocal with COLOR and a slow drift.
GRANULAR	Grains from the loaded sample (or a built-in chord bed). COLOR = grain size, DETUNE = scatter.
DUB WASH	Minor 7 chord, slow filter, chorus. The dub chord bed.
TAPE LOOP	Soft triangle/saw with wow, flutter, saturation and hiss.
NOISE BED	Noise through a resonant band-pass at the note: wind, air.
SAMPLE	Chromatic sampler from the analysed root note.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
CUTOFF M	0 % to 100 % default 55 %	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
RES	0 % to 100 % default 10 %	Resonance of the model's filter. On ACID 303-TYPE this is the squelch.
ENV AMT M	0 % to 100 % default 10 %	How far the filter envelope opens the filter on each note. Accented notes open further.
GATE	0 % to 100 % default 75 %	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
DENSITY M	0 % to 100 % default 0 %	Ghost notes. 0-50 %: one echo of every note fades in one ECHO SPACE later (1/8 hats become 1/16 hats with ECHO SPACE at 1/16). 50-100 %: two more ghosts at half that spacing (1/32 rolls). Each ghost is quieter, darker and shorter than the note before it. Ghosts are played softer and also faded in after the instrument's velocity response, so at the first notch they are 30 dB down: a fill really rises from silence. A quiet ghost never cuts the ringing kick or chokes the previous hit.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
DRIVE M	0 % to 100 % default 0 %	Saturation before the filter. Adds weight at low settings, grit at high.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

SETUP

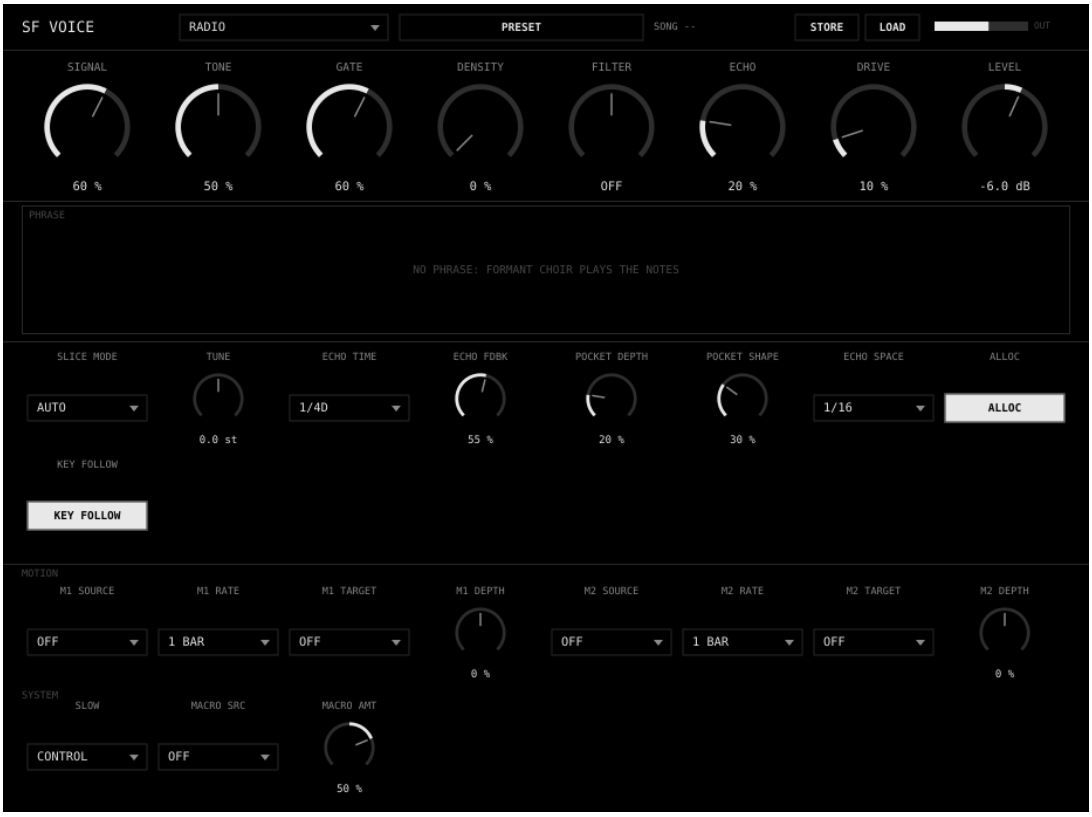
CONTROL	RANGE / DEFAULT	WHAT IT DOES
DECAY	0 % to 100 % default 50 %	Length of the filter envelope (and amp decay on most models). Short = plucky, long = sweeping. ACID: the classic decay knob.
ATTACK	0 % to 100 % default 0 %	Adds attack time to amp and filter envelopes. Up for swells.
ACCENT	0 % to 100 % default 50 %	How strongly accented notes (velocity above about 108) stand out: louder, brighter, shorter filter decay.
GLIDE	0 % to 100 % default 20 %	Portamento between notes (non-acid models). ACID slides only when notes overlap, like a 303.
TUNE	-24.0 st to 24.0 st default 0.0 st	Pitch offset in semitones (-24 to +24).
DETUNE	0 % to 100 % default 50 %	Spread between oscillators and unison voices. Also sets grain scatter on GRANULAR.
COLOR	0 % to 100 % default 50 %	Model-specific tone control. Mono subtractive: sub-oscillator level. ACID: below 50 % saw, above 50 % square. ORGAN: drawbar brightness. FM: modulation index. SYNC: sync sweep. WAVETABLE: table position (sine > saw > square > vocal). GRANULAR: grain size. NOISE BED: resonance of the noise.
LAYER MODEL	STRING MACHINE, POLY PAD, WAVETABLE, GRANULAR, DUB WASH, TAPE LOOP, NOISE BED, SAMPLE default POLY PAD	Second model played with every note, blended by LAYER.
LAYER	0 % to 100 % default 0 %	Blend of the LAYER MODEL under the main model. Use it to put a sub under a stab or an acid line on a bass.
CHORD	MODEL, NONE, MIN7, MIN9, SUS4, MAJ7 ... (9) default MODEL	Notes played per key. MODEL uses the model's own chord (STAB DUB = minor 7, STAB DETROIT = minor 9, ORGAN = minor triad). NONE forces single notes. Or choose MIN7, MIN9, SUS4, MAJ7, MIN, 5TH, OCT.
SPACE	0 % to 100 % default 50 %	Ensemble chorus amount, for the models that have one (strings, poly pad, stabs, hoover, organ).
POCKET DEPTH	0 % to 100 % default 30 %	How deep this module ducks around SF KICK (Pocket). 0 = no ducking.
POCKET SHAPE	0 % to 100 % default 25 %	One knob from transparent to pumping. Left: a narrow cut that follows the kick's pitch sweep (felt more than heard). Middle: the low band ducks for the kick's length. Right: full-band pump.
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO SWING	0 % to 100 % default 0 %	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: follows the KEY set on SF CONTROL (parts written in C play in the song's key). Off: ignores KEY.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF VOICE

INSTRUMENT, MIDI TRACK LATENCY 512 SAMPLES AT 48 KHZ

Spoken word and vocal phrases. Drop a phrase: it is sliced, and C3 plays slice 1, C#3 slice 2 and so on; keys below C3 play the whole phrase. Models treat it as radio, vocoder, robot or choir. Built-in dub echo. drosaka Voice stays separate.



USING IT

— Drop a phrase on the window (or LOAD). Markers show the slices and their keys.

— RADIO with SIGNAL low is a lost transmission; ride SIGNAL up to bring the voice in clear.

— ECHO is a send into a long dub echo (ECHO TIME, ECHO FDBK).

— With no phrase loaded, notes play a formant choir (TONE morphs the vowel).

MODELS

NAME	SOUND
SAMPLE	Plays the slices clean.
RADIO	Band-limited, driven, noisy: CB / transmission. SIGNAL = clarity.
VOCODER	16-band vocoder, saw carrier at the TONE pitch.
ROBOT	Vocoder with a pulse carrier at 8 kHz / 8-bit.
CHOIR	Vocoder with a minor-chord carrier.
STUTTER	Loops the start of each slice; TONE sets the loop length.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SIGNAL M	0 % to 100 % default 60 %	Clarity. RADIO: from lost to clear. Vocoders: sibilance and high-band brightness.
TONE	0 % to 100 % default 50 %	Model-specific: carrier pitch C1-C4 (vocoder, robot, choir), loop length (STUTTER), vowel (choir synth).
GATE	0 % to 100 % default 60 %	Slice length. Low values clip each slice short.
DENSITY M	0 % to 100 % default 0 %	Re-triggers the slice as ghost repeats (stutter fills).
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
ECHO M	0 % to 100 % default 20 %	Send into the built-in dub echo.
DRIVE	0 % to 100 % default 10 %	Saturation on the voice.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

SETUP

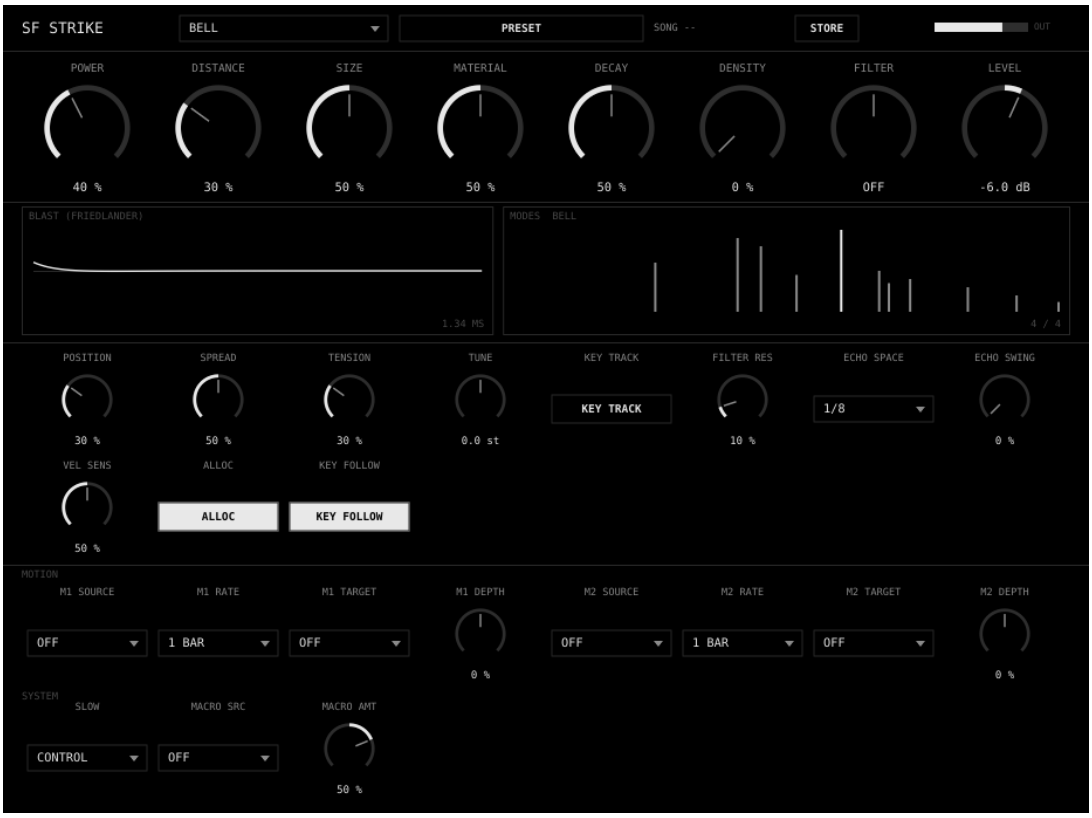
CONTROL	RANGE / DEFAULT	WHAT IT DOES
SLICE MODE	AUTO, EQUAL 8, EQUAL 16, WHOLE default AUTO	AUTO (detected onsets), EQUAL 8, EQUAL 16, WHOLE (every key plays the whole phrase).
TUNE	-12.0 st to 12.0 st default 0.0 st	Playback pitch (varispeed) in semitones.
ECHO TIME	1 BAR, 1/2, 1/4, 1/4D, 1/4T, 1/8 ... (9) default 1/4D	Echo time, synced (1 BAR to 1/16). Default 1/4 dotted.
ECHO FDBK	0 % to 100 % default 55 %	Echo feedback.
POCKET DEPTH	0 % to 100 % default 20 %	How deep this module ducks around SF KICK (Pocket). 0 = no ducking.
POCKET SHAPE	0 % to 100 % default 30 %	One knob from transparent to pumping. Left: a narrow cut that follows the kick's pitch sweep (felt more than heard). Middle: the low band ducks for the kick's length. Right: full-band pump.
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: playback and carriers follow the KEY set on SF CONTROL (phrases written in C play in the song's key). Off: ignores KEY.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF STRIKE

INSTRUMENT, MIDI TRACK

Blast-excited resonant objects. A pressure wave from a charge of POWER at DISTANCE strikes a bell, plate, bar, tube, membrane or gong, modelled as up to 16 resonant modes. Industrial impacts, distant detonations, metal that rings for bars.



USING IT

— Put it on a MIDI track. Any note strikes the object (KEY TRACK on: the note sets the pitch, C3 = TUNE).

— POWER is the charge: low is a tap, high a blast. Above about 70 % the object FRACTURES: its modes split and detune, debris rattles and it distorts. The display shows FRACTURE in amber.

— DISTANCE moves the blast away: quieter, longer and darker, exciting the low modes. Close and small is a sharp crack; far and big is a deep push.

— SIZE sets the pitch (bigger = lower). MATERIAL sets how long the high modes ring (wood to glass). DECAY sets the overall ring.

— DENSITY adds ghost strikes like on the other instruments; each ghost is a smaller charge.

— With a KEY set on SF CONTROL the object's main partial lands on the key root; SIZE then picks the octave.

OBJECTS

NAME	SOUND
BELL	Church-bell partials (hum, prime, tierce, quint, nominal). Long ring; the nominal is the pitch you hear.
PLATE	Sheet metal: dense pairs of close modes that beat. Thunder sheet, industrial clang.
BAR	Free bar (anvil, xylophone ratios). Short and pure.
TUBE	Pipe or tubular bell. Bright and long.
MEMBRANE	Drum head (circular membrane ratios). Short, thuddy.
GONG	Sixteen dense modes, the longest ring, and the strongest pitch glide after a hard hit (TENSION).

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
POWER M	0 % to 100 % default 40 %	Size of the charge. Velocity scales it (VEL SENS). Above about 70 % the object fractures.
DISTANCE	0 % to 100 % default 30 %	Stand-off. Further = quieter, longer, darker blast that excites fewer high modes.
SIZE	0 % to 100 % default 50 %	Object size, which sets the pitch: the fundamental goes from about 1.4 kHz (small) to 45 Hz (large).
MATERIAL	0 % to 100 % default 50 %	Wood (0) to glass (100 %): how long the high modes ring compared with the low ones.
DECAY M	0 % to 100 % default 50 %	Ring time, 0.15 to 9 s, times the object's own (a gong rings longer than a bar). Four strikes ring at once; a fifth fades the quietest out over 5 ms.
DENSITY M	0 % to 100 % default 0 %	Ghost notes. 0-50 %: one echo of every note fades in one ECHO SPACE later (1/8 hats become 1/16 hats with ECHO SPACE at 1/16). 50-100 %: two more ghosts at half that spacing (1/32 rolls). Each ghost is quieter, darker and shorter than the note before it. Ghosts are played softer and also faded in after the instrument's velocity response, so at the first notch they are 30 dB down: a fill really rises from silence. A quiet ghost never cuts the ringing kick or chokes the previous hit.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
POSITION	0 % to 100 % default 30 %	Strike point. Striking near a node mutes the modes that have one there, so it changes the colour.
SPREAD	0 % to 100 % default 50 %	Spreads the modes across the stereo field.
TENSION	0 % to 100 % default 30 %	How sharp a hard hit sounds before it settles (the pitch glides down after the blast).
TUNE	-24.0 st to 24.0 st default 0.0 st	Pitch in semitones (-24 to +24).
KEY TRACK	ON / OFF default OFF	On: the note number sets the pitch (C3 = TUNE). Off: every note strikes the same pitch.
FILTER RES	0 % to 100 % default 10 %	Resonance of the output FILTER.
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/8	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO SWING	0 % to 100 % default 0 %	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
VEL SENS	0 % to 100 % default 50 %	How much velocity scales the charge. 0 = every strike the same.
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: with a KEY set on SF CONTROL, the main partial lands on the nearest key root and TUNE moves from there. Off: ignores KEY.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF DECK

INSTRUMENT, MIDI TRACK

Two DJ decks inside the live set. Tracklist with background analysis (tempo, beat grid, loudness), tempo sync to Live with key lock, starts on phrase boundaries, hot cues from MIDI, slip loop roll, per-deck isolator and filter, crossfader, CUE output.



USING IT

— Drop tracks on the window. Each is analysed in the background (ANALYSING, then BPM and LUFS).

— Click A or B on a row to load it into a deck. X removes it from the list.

— PLAY A starts the deck on the next QUANT boundary (default 4 bars), locked to Live's tempo. PLAY off fades it out.

— Hot cues: C1-D#1 jump deck A to cue 1-4 on the next beat, C2-D#2 deck B. Default cues at beats 0, 32, 64, 96.

— Headphones: make an audio track with Audio From = the DECK track, choose CUE, route it to your headphone outputs, and turn CUE A or CUE B on.

— Every deck start is written to Documents/SF Live Suite/tracklog.txt.

— Decks follow Live's tempo. A plugin cannot change Live's tempo: use Live's tempo automation for tempo moves.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
XFADE	0 % to 100 % default 0 %	Crossfader, constant power. 0 = deck A, 100 % = deck B.
FILTER A	LP 100 to HP 100 default OFF	Deck A bipolar DJ filter.
FILTER B	LP 100 to HP 100 default OFF	Deck B bipolar DJ filter.
LOW A	0 % to 100 % default 75 %	Deck A low band: 0 = kill, 75 % = 0 dB, 100 % = +6 dB.
LOW B	0 % to 100 % default 75 %	Deck B low band.
ROLL	OFF, 1/4, 1/8, 1/16, 1/32 default OFF	Slip loop roll on the playing decks: 1/4 to 1/32. The track keeps running underneath; OFF returns to where it would have been.
GATE	0 % to 100 % default 0 %	1/16 trance gate depth on the deck mix.
LEVEL	-36.0 dB to 6.0 dB default -3.0 dB	Output level. Default is set so the module sits sensibly in a mix.

SETUP

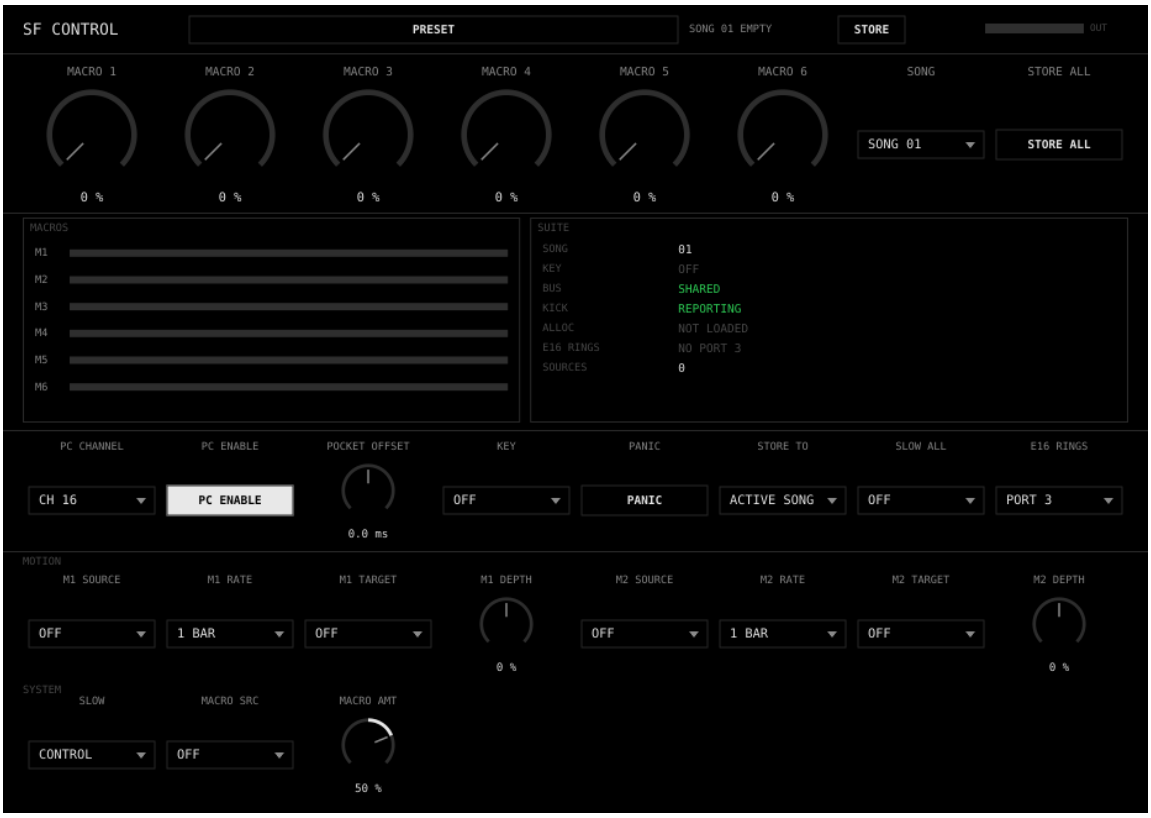
CONTROL	RANGE / DEFAULT	WHAT IT DOES
PLAY A	ON / OFF default OFF	Start (on the next QUANT boundary) / stop deck A.
PLAY B	ON / OFF default OFF	Start / stop deck B.
MID A	0 % to 100 % default 75 %	Deck A mid band.
HIGH A	0 % to 100 % default 75 %	Deck A high band.
MID B	0 % to 100 % default 75 %	Deck B mid band.
HIGH B	0 % to 100 % default 75 %	Deck B high band.
QUANT	1 BEAT, 1 BAR, 4 BAR, 8 BAR, 16 BAR default 4 BAR	Where PLAY starts: next beat, bar, 4, 8 or 16 bars.
KEY LOCK	ON / OFF default ON	On: time-stretch keeps the track's pitch when its tempo changes. Off: varispeed.
CUE A	ON / OFF default OFF	Send deck A to the CUE output (pre-crossfader).
CUE B	ON / OFF default OFF	Send deck B to the CUE output.
TRIM TARGET	-16.0 LUFS to -6.0 LUFS default -10.0 LUFS	Every track is trimmed to this loudness (LUFS) so decks match.
FILTER RES	0 % to 100 % default 20 %	Resonance of both deck filters.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF CONTROL

INSTRUMENT, MIDI TRACK

The suite's brain. Six macros every SF plugin can follow, song select (Program Change), STORE ALL and STORE TO, and SLOW ALL. Makes no sound.



USING IT

— Put it on its own MIDI track receiving the OXI One.

— Program Change on channel 16 selects the song. AU: in any Live version. VST3: Live 12.2 and later deliver it as a program change (SF CONTROL lists SONG 01-16); older versions don't.

— Every plugin changes on the same bar line, whatever order Live processes the tracks in. A change sent within 30 ms after a bar line (on the downbeat) counts for that bar; later in the bar, for the next one.

— To save a song: set every plugin and press STORE ALL. It goes into the song in force (SONG).

— To save into another song, choose it in STORE TO first, then press STORE ALL. That song isn't recalled, so nothing you hear changes; it becomes the song in force (SONG follows) and STORE TO goes back to ACTIVE SONG. If a song change is waiting for its bar, that change still lands and stays the song in force.

— The display shows macro levels, the song, the key, whether the bus is shared, SF KICK is reporting and SF ALLOC is running, and the E16 RINGS port.

— KEY: write parts in C and set each song's key here. Every pitched SF instrument follows; KICK, RUMBLE and STRIKE tune themselves to the key root.

— PANIC: map it to a button you can find in the dark. Every SF plugin fades out over 5 ms, clears every voice, tail and delay line, and stays silent while you hold it.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
MACRO 1	0 % to 100 % default 0 %	Macro 1. Any SF plugin with MACRO SRC = MACRO 1 follows it.
MACRO 2	0 % to 100 % default 0 %	Macro 2.
MACRO 3	0 % to 100 % default 0 %	Macro 3.
MACRO 4	0 % to 100 % default 0 %	Macro 4.
MACRO 5	0 % to 100 % default 0 %	Macro 5.
MACRO 6	0 % to 100 % default 0 %	Macro 6.
SONG	SONG 01, SONG 02, SONG 03, SONG 04, SONG 05, SONG 06 ... (16) default SONG 01	Selects the song (01-16). Every SF plugin recalls its stored values on the next bar. A recalled knob is yours again the moment you turn it.
STORE ALL	HOLD	Stores every SF plugin's current values into the song STORE TO names: ACTIVE SONG, the song in force, unless you chose another.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
PC CHANNEL	CH 1, CH 2, CH 3, CH 4, CH 5, CH 6 ... (16) default CH 16	MIDI channel that Program Change is read from (default 16).
PC ENABLE	ON / OFF default ON	Turns Program Change song select on or off.
POCKET OFFSET	-20.0 ms to 20 ms default 0.0 ms	Timing calibration for Pocket and RUMBLE, -20 to +20 ms (+ = later). Set once, by ear, if the pump lands early or late in your Live setup; leave at 0 otherwise.
KEY	OFF, C, C#, D, D#, E ... (13) default OFF	Key of the song: OFF, C ... B. Parts written in C play in this key on every SF instrument with KEY FOLLOW on (nearest transposition, -6 to +5 semitones). Stored per song, it changes on the song's bar; sounding notes follow. Only while SF CONTROL runs.
PANIC	HOLD	Hold: every SF plugin fades out over 5 ms, clears every voice, tail and delay line and stays silent (no input, no new notes) until you let go; every header shows PANIC in red. Release: back in over 5 ms; DECK keeps its place in time. A set never reopens with PANIC held.
STORE TO	ACTIVE SONG, SONG 01, SONG 02, SONG 03, SONG 04, SONG 05 ... (17) default ACTIVE SONG	Where STORE ALL stores: ACTIVE SONG (the song in force) or SONG 01-16. Another song isn't recalled: what plays now goes in, that song becomes the song in force (SONG follows) and STORE TO goes back to ACTIVE SONG.
SLOW ALL	OFF, 1 BAR, 2 BAR, 4 BAR, 8 BAR, 16 BAR, 32 BAR, 64 BAR default OFF	How long your next knob moves take (OFF, 1-64 BARS) in every SF plugin whose SLOW is CONTROL. A move keeps its length: turning SLOW ALL never touches a travelling knob. Song recalls follow it. Only while SF CONTROL runs.
E16 RINGS	OFF, PORT 1, PORT 2, PORT 3 default PORT 3	E16 port for GLIDE RINGS (OFF, PORT 1-3): in the SF SHOW scene a travelling knob's ring turns green. Display: PORT 3 in green once found.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

9 EFFECTS

SF STRIP

AUDIO EFFECT LATENCY 512 SAMPLES AT 48 KHZ

The suite's performance row for any track. Put it after any Live instrument, sampler, drum rack or hardware input: the track gets DENSITY, GATE, FILTER, DRIVE, POCKET and a SPACE send, joins ALLOC, and follows SLOW, macros and songs like every SF plugin.



USING IT

— Last device on the track (after the instrument and anything else).

— Live sends no MIDI to audio effects, so STRIP works on the hits it hears. GATE closes each detected hit after a set length (staccato to held); DENSITY adds darker, quieter echo ghosts between hits.

— Set ROLE so ALLOC knows the track's priority (drum rack = PERC, a bass synth = BASS).

— THROW sends the track into its own SPACE (dub echo or reverb) while held.

— On a long, sustained sound GATE and DENSITY act on its attacks only; SENSITIVITY sets how small an attack counts as a hit.

MODELS

NAME	SOUND
Chain	GATE > DENSITY > DRIVE > FILTER > SPACE > POCKET > ALLOC > LEVEL (fixed order).

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
DENSITY M	0 % to 100 % default 0 %	Ghost echoes of each hit: 0-50 % one echo one ECHO SPACE later, 50-100 % two more at half spacing. Each darker and quieter.
GATE M	0 % to 100 % default 100 %	GATE MODE HITS: how long each detected hit stays open, from 1/64 beat at 0 % to 2 beats; 97 % and up = never closes. PATTERN: how long each pattern step stays open.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
DRIVE M	0 % to 100 % default 0 %	Saturation, 0 to +30 dB into the shaper (CHARACTER sets tape to fuzz), 4x oversampled, fading in from clean.
POCKET	0 % to 100 % default 0 %	Pocket depth: ducks the track around SF KICK.
SPACE M	0 % to 100 % default 0 %	Send into the strip's own SPACE (dub echo or reverb). The reverb rings out when the send closes.
THROW	HOLD	Hold: full send into SPACE.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

SETUP

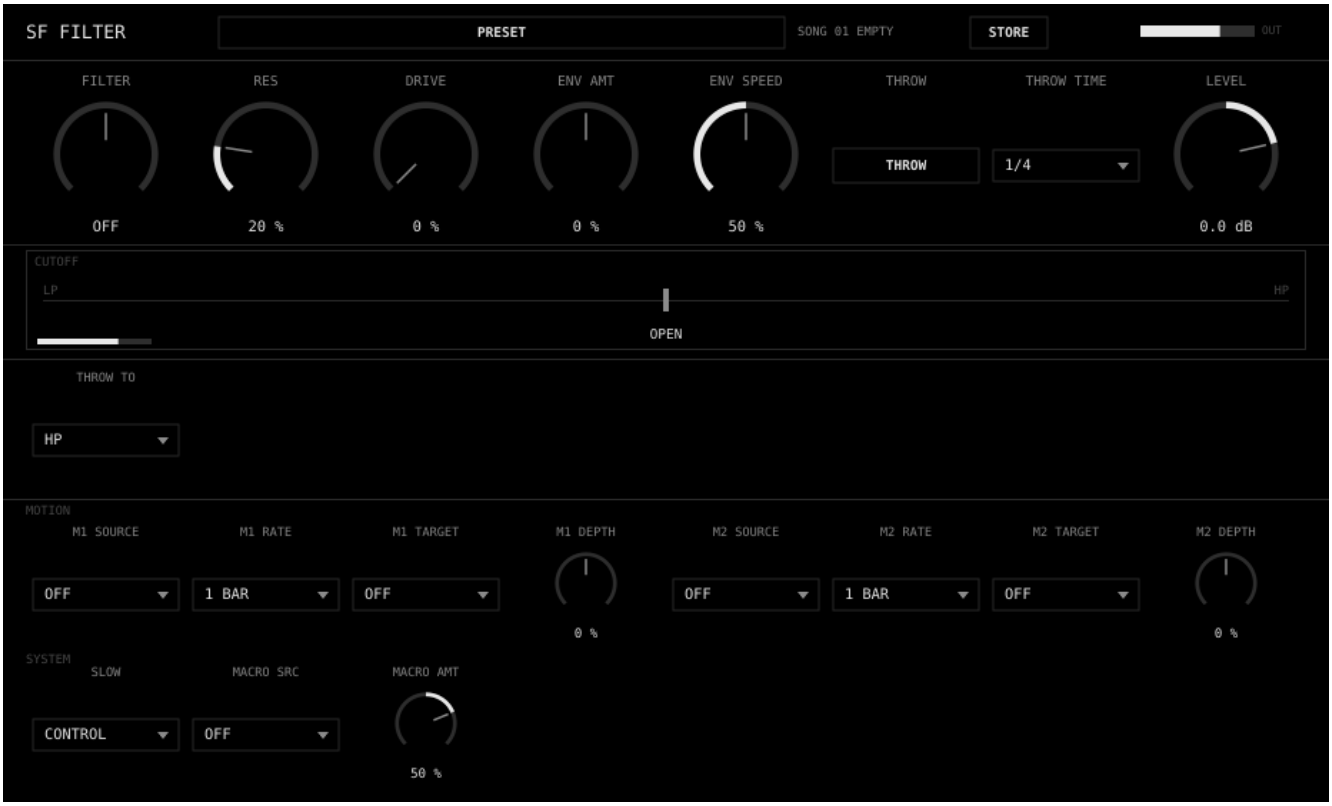
CONTROL	RANGE / DEFAULT	WHAT IT DOES
ROLE	KICK, BASS, VOICE, LEAD, PERC, RUMBLE ... (9) default OTHER	Priority in ALLOC and the name shown (STRIP BASS, STRIP PERC ...).
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO PROB	0 % to 100 % default 100 %	Chance each hit gets its echoes.
DARKEN	0 % to 100 % default 40 %	How much darker each generation of echoes is.
GATE MODE	HITS, PATTERN default HITS	HITS (closes each detected hit) or PATTERN (rhythmic gate).
PATTERN	FULL, 8THS, OFFBEAT, OFF 8THS, TRESILLO, GALLOP ... (12) default 8THS	Gate pattern for PATTERN mode.
RATE	1/8, 1/16, 1/32 default 1/16	Step size for PATTERN mode: 1/8, 1/16, 1/32.
CHARACTER	0 % to 100 % default 0 %	Drive character: tape (0) to fuzz (100 %).
FILTER RES	0 % to 100 % default 20 %	Resonance of the output FILTER. Higher gives a sharper, more vocal sweep.
POCKET SHAPE	0 % to 100 % default 30 %	One knob from transparent to pumping. Left: a narrow cut that follows the kick's pitch sweep (felt more than heard). Middle: the low band ducks for the kick's length. Right: full-band pump.
SPACE TYPE	REVERB, DUB ECHO default DUB ECHO	REVERB or DUB ECHO (ping-pong, 1/8 to 3/16).
SPACE SIZE	0 % to 100 % default 50 %	Reverb size and decay / echo time and feedback.
SENSITIVITY	0 % to 100 % default 50 %	How small an attack still counts as a hit (GATE HITS and ECHO PROB).

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF FILTER

AUDIO EFFECT

Bipolar performance filter with drive, envelope follow and a throw.



USING IT

— One knob from low-pass through open to high-pass.

— Hold THROW for a timed sweep to the extreme (THROW TIME); release and it returns.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
FILTER M	LP 100 to HP 100 default OFF	Bipolar 24 dB/oct filter. Centre open, left low-pass, right high-pass.
RES	0 % to 100 % default 20 %	Resonance, level-compensated.
DRIVE	0 % to 100 % default 0 %	Saturation before the filter, 2x oversampled. Fades in from exactly clean, so the first notch never jumps the level.
ENV AMT	-100 % to 100 % default 0 %	Envelope follow: the input's level pushes the cutoff up (+) or down (-).
ENV SPEED	0 % to 100 % default 50 %	Speed of the envelope follower.
THROW	HOLD	Hold: the filter travels to the extreme over THROW TIME. Release: back in a quarter of that.
THROW TIME	4 BAR, 2 BAR, 1 BAR, 1/2, 1/4, 1/4D ... (11) default 1/4	Length of the throw: 4 BAR to 1/16.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

SETUP

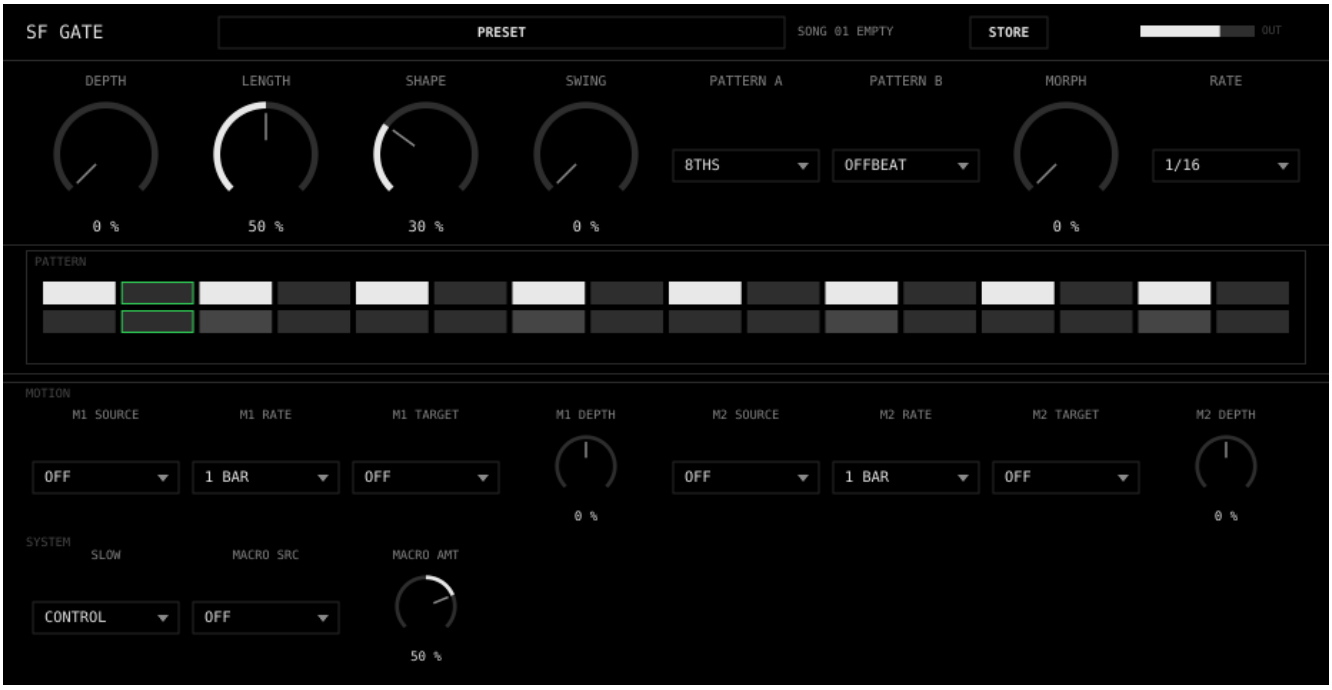
CONTROL	RANGE / DEFAULT	WHAT IT DOES
THROW TO	HP, LP default HP	Direction of the throw: HP or LP.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF GATE

AUDIO EFFECT

Rhythmic gate with 16 patterns, morphing between two of them.



USING IT

— The tool that turns a held note into a rhythm.

— Set PATTERN A and B, then ride MORPH between them; DEPTH sets how hard the gate cuts.

REFERENCE

NAME	SOUND
Patterns	FULL, 8THS, OFFBEAT, OFF 8THS, TRESILLO, GALLOP, SKIP 1, SKIP 2, STAB, PUMP, BROKEN, CLAVE, 3-3-2, HALF, SPARSE, DOTTED.

PERFORMANCE ROW

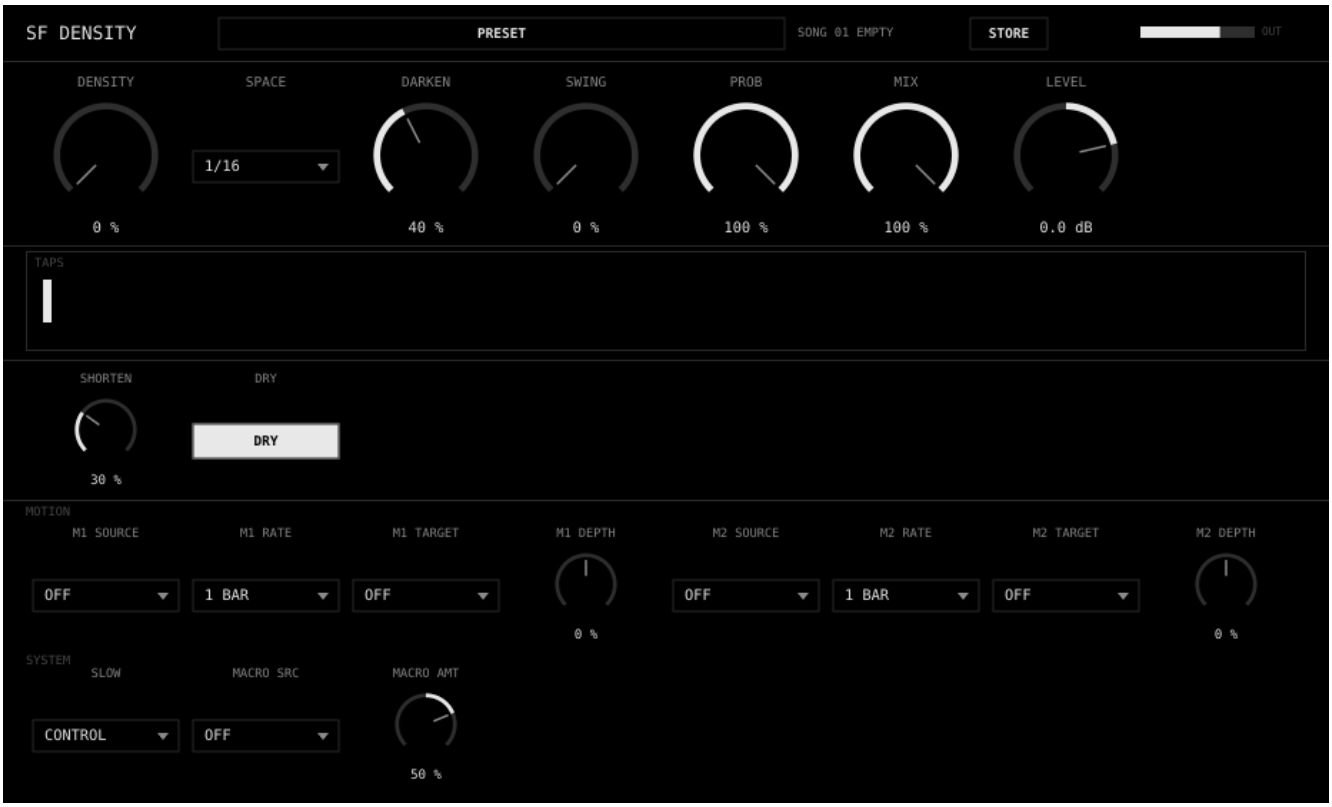
CONTROL	RANGE / DEFAULT	WHAT IT DOES
DEPTH M	0 % to 100 % default 0 %	How far the gate closes. 0 = no effect.
LENGTH	0 % to 100 % default 50 %	How long each step stays open.
SHAPE	0 % to 100 % default 30 %	Edges: hard (clicky) to soft (swelling).
SWING	0 % to 100 % default 0 %	Delays every second step.
PATTERN A	FULL, 8THS, OFFBEAT, OFF 8THS, TRESILLO, GALLOP ... (16) default 8THS	First pattern.
PATTERN B	FULL, 8THS, OFFBEAT, OFF 8THS, TRESILLO, GALLOP ... (16) default OFFBEAT	Second pattern.
MORPH	0 % to 100 % default 0 %	Blend from PATTERN A to PATTERN B. In between, steps only in one pattern play partly open.
RATE	1/8, 1/16, 1/32, 1/16T default 1/16	Step size: 1/8, 1/16, 1/32, 1/16T.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF DENSITY

AUDIO EFFECT

The send-delay hat trick as a plugin: fills sparse rhythms with ghost echoes, each generation darker and quieter.



USING IT

— Put it on hats or percussion (insert, or a return with DRY off).

— DENSITY 50 % with SPACE 1/16 turns 1/8 hats into 1/16s; above 50 % adds 1/32 ghosts.

— PROB decides, per incoming hit, whether its echoes play.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
DENSITY M	0 % to 100 % default 0 %	0-50 %: one echo one SPACE later. 50-100 %: two more ghosts at half spacing.
SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Echo spacing: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D.
DARKEN	0 % to 100 % default 40 %	How much darker each generation of echoes is.
SWING	0 % to 100 % default 0 %	Delays the first echo.
PROB	0 % to 100 % default 100 %	Chance each incoming hit gets its echoes.
MIX	0 % to 100 % default 100 %	Echo level.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SHORTEN	0 % to 100 % default 30 %	Echoes carry only the start of each hit (tighter ghosts).
DRY	ON / OFF default ON	Passes the input. Turn it off on a return track, so the return gives only the echoes. Not stored in songs or presets.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF DELAY

AUDIO EFFECT LATENCY 512 SAMPLES AT 48 KHZ

Tape-style dub delay. Changing TIME bends the pitch as the tape head moves; feedback above 100 % saturates instead of exploding; THROW and FREEZE for live moves. Ducks its repeats around the kick.



USING IT

— As a return (DRY off): INPUT THROW ONLY and THROW mapped to a button gives dub throws.

— Turn TIME during a tail for the tape-bend. SLEW sets how long the bend takes.

— FEEDBACK past 100 % with TONE swept = the runaway dub effect, safe.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
TIME	1 BAR, 1/2, 1/4, 1/4D, 1/4T, 1/8 ... (14) default 1/8D	Delay time, synced: 1 BAR to 1/32, then 5/16 and 7/16 (odd lengths: the repeats walk across the beat). Changing it slews the head (pitch bend).
FEEDBACK M	0 % to 115 % default 45 %	Repeats, up to 115 %. Above 100 % the loop saturates instead of growing.
TONE	LP 100 to HP 100 default OFF	Bipolar filter inside the feedback loop: left darker, right thinner. Each repeat goes further than the last.
DRIVE	0 % to 100 % default 20 %	Saturation inside the feedback loop (2x oversampled): each repeat dirtier than the last.
THROW	HOLD	Hold to send into the delay (louder send in INPUT ALWAYS, the only send in THROW ONLY).
FREEZE	ON / OFF default OFF	Loops the current repeats forever, losslessly: no feedback loss, no filtering, the tape wow stops (10 ms crossfade in and out); input muted. Turning TIME while frozen bends the loop (a tape-speed move) and it locks again when the head settles.
MIX M	0 % to 100 % default 30 %	Wet level.
WIDTH	0 % to 100 % default 50 %	Stereo width of the repeats.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SLEW	10 ms to 3.00 s default 400 ms	How long a TIME change takes (the length of the pitch bend).
MODE	STEREO, PING-PONG default PING-PONG	STEREO or PING-PONG.
INPUT	ALWAYS, THROW ONLY default ALWAYS	ALWAYS, or THROW ONLY (nothing enters except while THROW is held).
DUCK	0 % to 100 % default 0 %	Pocket on the repeats: they duck under the kick.
DUCK SHAPE	0 % to 100 % default 60 %	Pocket shape for the repeats.
WOW	0 % to 100 % default 10 %	Tape wow on the repeats.
LO CUT	20 Hz to 600 Hz default 80 Hz	Low cut inside the loop.
DRY	ON / OFF default ON	Passes the input at unity (MIX sets only the repeats). Turn it off on a return track. Not stored in songs or presets.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF REVERB

AUDIO EFFECT LATENCY 512 SAMPLES AT 48 KHZ

Dark feedback-network reverb with freeze, throw, gated mode and ducking (kick pocket plus dry-follow, so tails bloom in the gaps).



USING IT

— DUCK at 30-60 % keeps big spaces from washing out the groove.

— FREEZE holds the space as a drone; THROW sends a single hit.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SIZE M	0 % to 100 % default 60 %	Room size.
DECAY M	0.30 s to 30.00 s default 4.00 s	Decay time of the lows, 0.3 to 30 s (measured: DECAY is the real RT60).
DARK	0 % to 100 % default 50 %	Damping: shortens the decay of the highs only, so the tail gets darker without getting shorter.
PREDELAY	0, 1/64, 1/32, 1/16, 1/8, 1/4 default 1/32	Synced predelay: 0, 1/64, 1/32, 1/16, 1/8, 1/4.
DUCK	0 % to 100 % default 30 %	Ducks the tail under the kick and while the dry signal plays.
FREEZE	ON / OFF default OFF	Infinite hold: the space keeps its level and brightness (measured over 14 s). The input fades into the frozen space over 10 ms, then is muted.
THROW	HOLD	Hold to send (or the only send in INPUT THROW ONLY).
MIX M	0 % to 100 % default 25 %	Wet level.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
MOD	0 % to 100 % default 30 %	Modulation of the tail (chorus-like movement).
WIDTH	0 % to 100 % default 80 %	Stereo width.
GATED	ON / OFF default OFF	Gated reverb: the tail is cut GATE TIME after each hit.
GATE TIME	1/16, 1/8, 1/4, 1/2 default 1/8	1/16, 1/8, 1/4, 1/2.
LO CUT	20 Hz to 800 Hz default 120 Hz	Low cut on the input.
DUCK SHAPE	0 % to 100 % default 50 %	Pocket shape for the tail.
INPUT	ALWAYS, THROW ONLY default ALWAYS	ALWAYS or THROW ONLY.
DRY	ON / OFF default ON	Passes the input at unity (MIX sets only the tail). Turn it off on a return track. Not stored in songs or presets.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF TERROIR

AUDIO EFFECT LATENCY 512 SAMPLES AT 48 KHZ

Convolution reverb for real places. Eight PLACE slots, each a space you captured (a clap, a balloon, or the sweep file played from a speaker and recorded) or any IR file, kept inside the Live set and recalled with songs. Its first eight controls are SF REVERB's, so it swaps in on a return.



USING IT

— Tie a song to a place: choose its PLACE (the boxes on the display), then STORE the song. A new place crossfades in with the tail it already has, never cut off.

— On B SPACE instead of SF REVERB: the E16's SPACE page fits as it is. Map SIZE, DECAY, DARK, DUCK and the two presses to it once (Cmd-M).

— LENGTH 1/4 or 1/8 turns a place into a gated hit in time; REVERSE with LENGTH 1/2 gives a swell that lands on the next beat.

— Stereo from one microphone: record the same sweep or clap from two spots (or a shotgun aimed at two parts of the room), name the files ... L and ... R, and drop both at once. Each take lines up at its own onset; their tails differ, so the place is wide (WIDTH under 100 % if too wide).

— The room you film in, on the whole mix: capture it from the camera's place with the speaker where you play, then put SF TERROIR on the recording (DRY on; preset MIX/IN THE ROOM): the sound sits in the room the picture shows.

CAPTURING A PLACE

SOURCE	HOW
CLAP	Clap once (or burst a balloon) and record a few seconds of the room. Drop the recording: the loudest event is used, from 1 ms before its onset to where its tail meets the noise, cut before a second clap. Quicker than the sweep, but the noise takes its tail sooner: a good backup.
SWEEP	SWEEP FILE (on the display) writes SF TERROIR SWEEP.wav (Music/SF Live Suite/TERROIR): 43 s, three 10 s sweeps. Play it from any speaker (a phone into a MiniRig is fine; by cable rather than Bluetooth if you can), record all of it (32-bit float, or under full scale: a clipped take blurs the place), drop the recording. The sweeps are found, the speaker's clock drift measured and corrected, the three averaged: the cleanest IR, and the speaker's distortion left out.
IR FILE	Any impulse response (WAV, AIFF, FLAC), mono or stereo, any rate. Up to 12 s is kept.
THE RESULT	Every place is scaled to the same energy (near SF REVERB's level), high-passed at 20 Hz and stored in the set as FLAC. The status line gives its kind, length, RT60 and how far it rises over the noise (under 30 dB: amber, try again closer or louder). A stereo take keeps the time and level difference between its mics. From a recorder's poly file the mics are used: tracks named MIX (a MixPre's MixL and MixR) are skipped, then the first two. Two channels carrying the same signal make a mono place.

IN THE FIELD

STEP	WHAT
MICS	Two mics recorded at once keep the room's own width. Cardioids in ORTF (capsules 17 cm apart, 110° between them) sound natural on headphones; 40 cm to 1 m apart, bigger. Pointed away from the speaker they hear more room and less speaker. Omnis: 40 cm to 1 m apart. A shotgun (a 416) gives an aimed, coloured place; one mic makes stereo in two takes (USING IT). Pads at 0, low cut off.
SPEAKER	Where the sound should come from, aimed at a wall or into the space. A small one is enough: the lows it can't play are lows a techno reverb cuts anyway. A bigger one plays louder, so a long tail stays over the noise longer (10 dB more: a sixth of the decay). Avoid one that adapts its sound as it plays; bass boost off.
PLAYER	Play the sweep file from the phone's file app, not a music app (no normalising, spatial audio or EQ), in airplane mode. By cable if you can; Bluetooth works (its clock is corrected), but a dropout spoils a sweep. As loud as the speaker goes without buzzing.
RECORDER	48 kHz, 32-bit float. The same gain on both mics, low cut and limiters off. Input 1 is the mic on your left as you face the speaker. The mix and its pan don't matter: SF TERROIR reads the mic tracks.
THE TAKE	Start recording, then press play. Keep still and quiet for the 43 s; stop a few seconds after the file ends. Take a second spot deeper in the room: there are eight slots. A minute of the place's own sound makes an ambience bed for a film.
BACK HOME	Name the file after the place (the slot takes its name) and drop it on a PLACE slot. The status line should read SWEEP x3 (from a poly file, FROM and the two mic tracks) and 30 dB or more over the noise. A sweep take that comes back as IMPULSE wasn't found: play it louder or closer.
IN THE MIX	START at 5-20 ms drops the speaker's direct sound and leaves the room. WIDTH under 100 % if two takes from one mic come out too wide; DARK if the mics or the speaker made the place bright.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SIZE M	50 % to 200 % default 100 %	The same place bigger (slower and lower) or smaller: the IR resampled, 50-200 %. While it moves, a new IR is built at most every 40 ms and crossfaded in.
DECAY M	10 % to 100 % default 100 %	Shortens the tail: the RT60 times DECAY. 100 % = as captured.
DARK	0 % to 100 % default 0 %	The tail darkens as it goes, up to 8 octaves a second at 100 %. 0 = as captured.
PREDELAY	0, 1/64, 1/32, 1/16, 1/8, 1/4 default 0	Synced predelay: 0, 1/64, 1/32, 1/16, 1/8, 1/4.
DUCK	0 % to 100 % default 30 %	Ducks the tail under the kick and while the dry signal plays.
FREEZE	ON / OFF default OFF	Holds the wet sound's spectrum as an endless texture (in over 150 ms, out over 400 ms) while the tail dies away; the input is muted while it holds.
THROW	HOLD	Hold to send (or the only send in INPUT THROW ONLY).
MIX M	0 % to 100 % default 25 %	Wet level.

SETUP

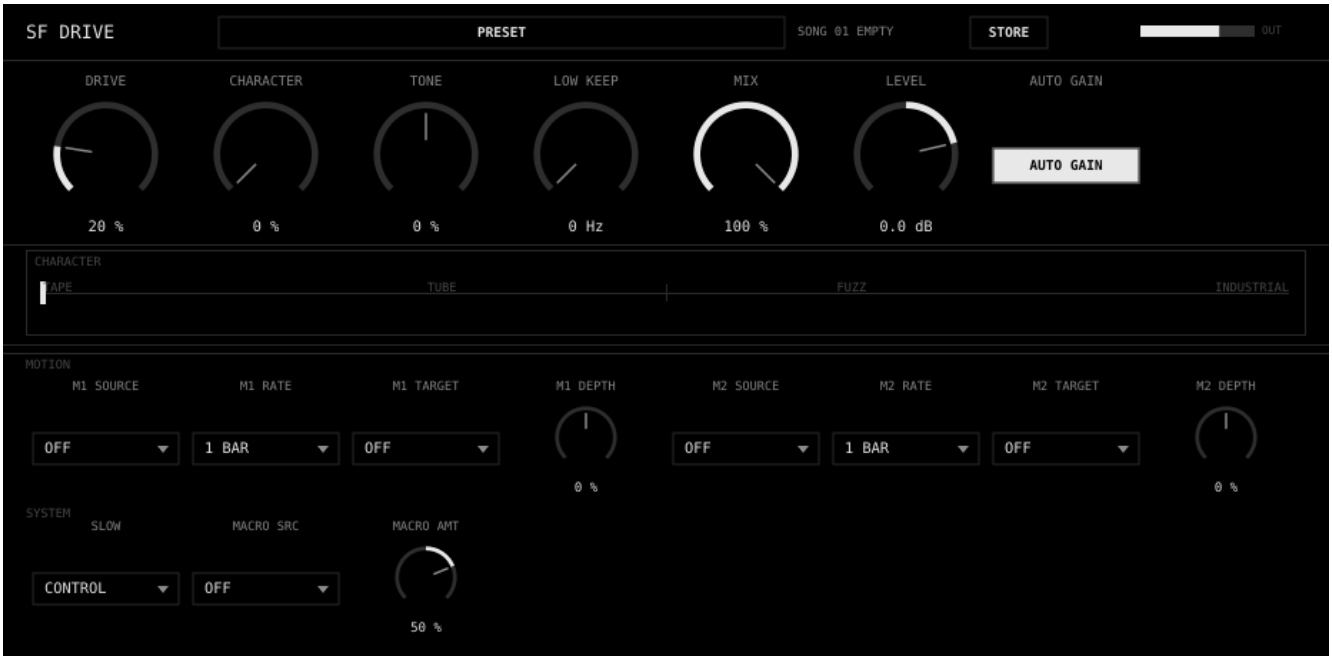
CONTROL	RANGE / DEFAULT	WHAT IT DOES
PLACE	1, 2, 3, 4, 5, 6, 7, 8 default 1	Slot 1-8 (or click the boxes on the display). An empty slot plays the built-in room. Stored in songs; presets never change it.
LENGTH	FULL, 4 BAR, 2 BAR, 1 BAR, 1/2, 1/4, 1/8, 1/16 default FULL	Cuts the IR to a note value (FULL, 4 BAR to 1/16), with a raised-cosine end.
REVERSE	ON / OFF default OFF	Plays the IR backwards: a swell that ends on the hit, LENGTH long.
START	0.0 ms to 250 ms default 0.0 ms	Skips the start of the IR, 0-250 ms: the direct sound and early reflections go, the tail stays.
WIDTH	0 % to 200 % default 100 %	Stereo width of the wet signal, 0-200 %. A mono capture is spread by a decorrelated copy of its tail.
LO CUT	20 Hz to 800 Hz default 120 Hz	Low cut on the input.
DUCK SHAPE	0 % to 100 % default 50 %	Pocket shape for the tail.
INPUT	ALWAYS, THROW ONLY default ALWAYS	ALWAYS or THROW ONLY.
DRY	ON / OFF default ON	Passes the input at unity (MIX sets only the tail). Turn it off on a return track. Not stored in songs or presets.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF DRIVE

AUDIO EFFECT LATENCY 5 SAMPLES AT 48 KHZ

Saturation with one CHARACTER knob from smooth to fuzz: TAPE, TUBE, FUZZ, INDUSTRIAL. 4x oversampled, auto gain.



USING IT

— Your smooth-then-heavy arc on a single knob: automate CHARACTER across a track.

— LOW KEEP keeps the sub clean while the rest distorts.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
DRIVE M	0 % to 100 % default 20 %	Amount, 0 to +36 dB into the shaper.
CHARACTER M	0 % to 100 % default 0 %	TAPE > TUBE > FUZZ > INDUSTRIAL (wavefold).
TONE	-100 % to 100 % default 0 %	Tilt EQ around 800 Hz.
LOW KEEP	0 Hz to 300 Hz default 0 Hz	Below this frequency the signal stays clean. 0 = off.
MIX	0 % to 100 % default 100 %	Dry/wet. The dry signal runs through an identical oversampler, so the blend is phase-aligned (no comb filtering) and MIX 0 is the clean input.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.
AUTO GAIN	ON / OFF default ON	Holds the driven signal's loudness near the input's, before MIX.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF RADIO

AUDIO EFFECT

Transmission processor: band-limit, dirt, noise, dropouts and squelch. SIGNAL is one knob from lost to clear: it also takes CRUSH and RATE from their setting at 0 to clean at 100 %.



USING IT

— On a voice, ride SIGNAL from 0 to 100 % to bring a ghost transmission into focus.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SIGNAL M	0 % to 100 % default 50 %	Clarity: band width, noise, dropouts, distortion, CRUSH and RATE all follow it.
BAND	300 Hz to 3.00 kHz default 1.20 kHz	Centre of the pass band.
WIDTH	0 % to 100 % default 50 %	Width of the pass band.
NOISE	0 % to 100 % default 30 %	Noise floor.
DROPOUT	0 % to 100 % default 20 %	How often the signal drops out (more at low SIGNAL). Each drop fades out and back in over about 10 ms.
SQUELCH	0 % to 100 % default 40 %	A squeal as each dropout ends: a whistle sliding down from the top of the band, a little breath around it, 0.1-0.2 s (longer with more SQUELCH). Fades in and out smoothly: no click.
RING	0 % to 100 % default 0 %	Ring modulation (47 Hz) for a metallic voice.
MIX	0 % to 100 % default 100 %	Dry/wet.

SETUP

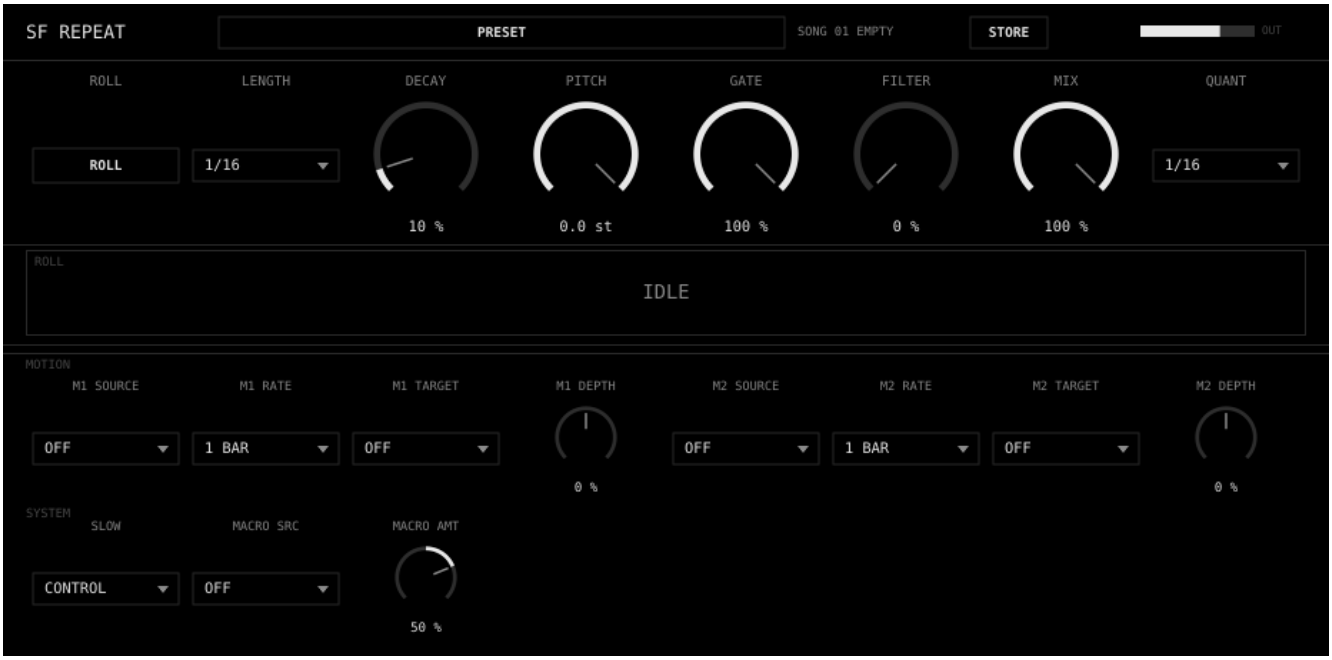
CONTROL	RANGE / DEFAULT	WHAT IT DOES
CRUSH	4.00 bit to 16.00 bit default 16.00 bit	Bit depth at SIGNAL 0; SIGNAL brings it back to 16 bits (16 = off).
RATE	2.00 kHz to 48.00 kHz default 48.00 kHz	Sample rate at SIGNAL 0; SIGNAL brings it back to full rate (48 kHz = off).

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF REPEAT

AUDIO EFFECT

Beat repeat for fills and builds. Hold ROLL: it captures LENGTH of audio from the next QUANT point and loops it, each pass quieter, lower, darker.



USING IT

— Map ROLL to a button; LENGTH 1/16 then 1/32 into a drop.

— PITCH negative = the classic falling roll.

PERFORMANCE ROW

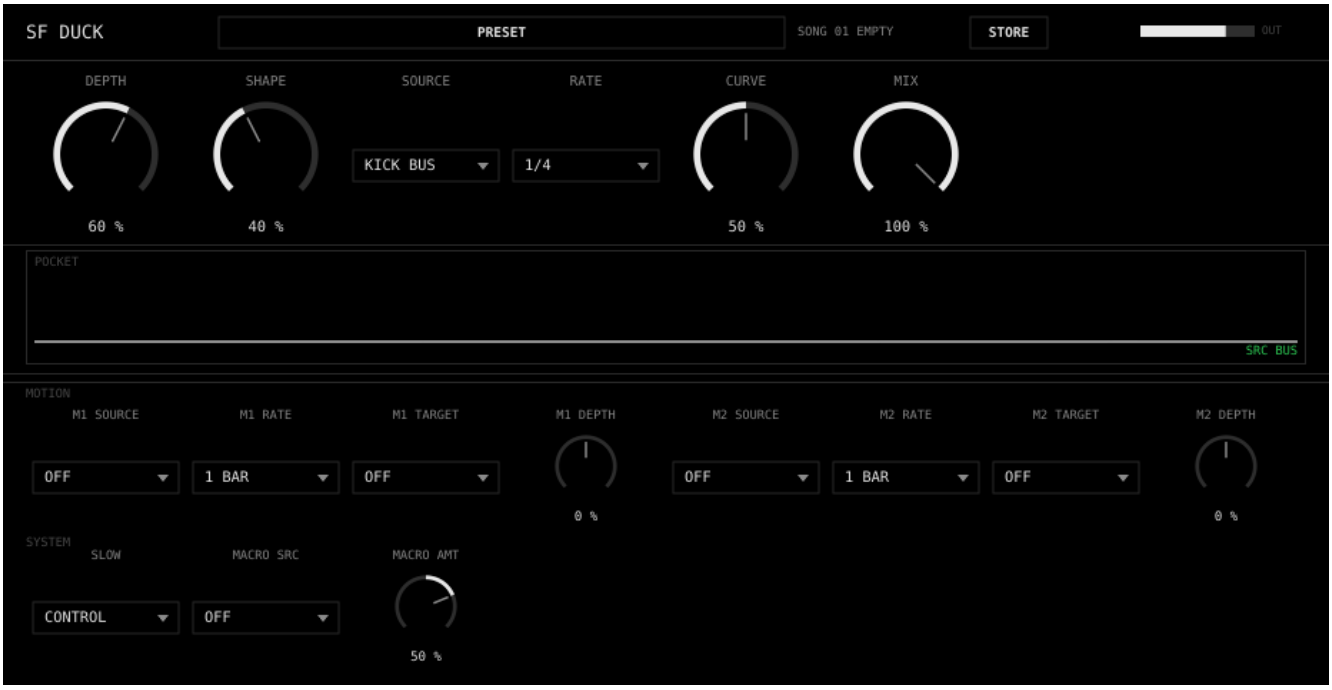
CONTROL	RANGE / DEFAULT	WHAT IT DOES
ROLL	HOLD	Hold to roll. The slice is captured from the QUANT point (up to 4 s) and kept for the whole hold. Release: the loop keeps playing while it crossfades back to the input.
LENGTH	1/4, 1/8, 1/16, 1/32, 1/8T, 1/16T default 1/16	Loop length: 1/4 to 1/32 and triplets.
DECAY	0 % to 100 % default 10 %	Level loss per pass.
PITCH	-12.0 st to 0.0 st default 0.0 st	Pitch drop over the repeats: falls by this amount every four passes.
GATE	0 % to 100 % default 100 %	How much of each pass plays.
FILTER	0 % to 100 % default 0 %	Darkens over the repeats.
MIX	0 % to 100 % default 100 %	Roll level over the input.
QUANT	OFF, 1/16, 1/8, 1/4 default 1/16	Start point: OFF (at once), 1/16, 1/8, 1/4.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF DUCK

AUDIO EFFECT LATENCY 512 SAMPLES AT 48 KHZ

Pocket as an effect for any audio: follows SF KICK on the bus, a sidechain input, or a tempo-synced volume shape.



USING IT

— Put it on any track that should pump with the kick: no sidechain routing needed.

— SOURCE SIDECHAIN for an external kick; GRID for a classic volume-shaper pump.

PERFORMANCE ROW

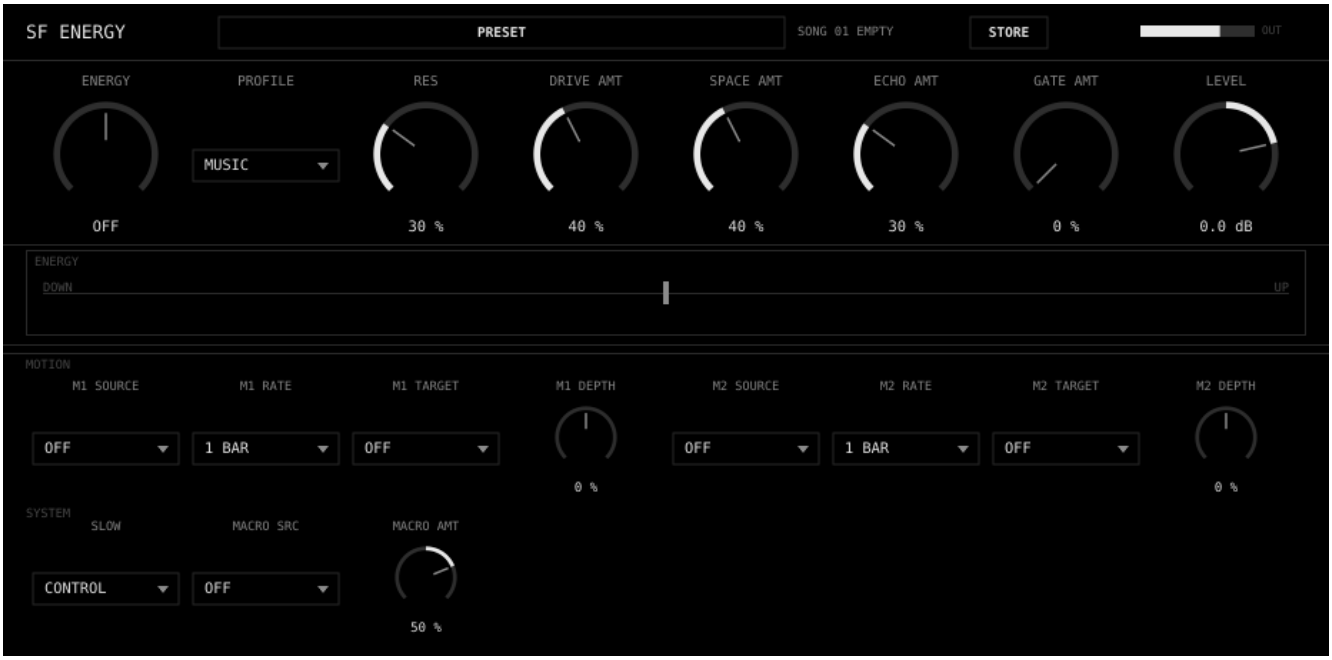
CONTROL	RANGE / DEFAULT	WHAT IT DOES
DEPTH M	0 % to 100 % default 60 %	How far it ducks.
SHAPE M	0 % to 100 % default 40 %	Transparent (pitch-tracking) to full pump.
SOURCE	KICK BUS, SIDECHAIN, GRID default KICK BUS	KICK BUS (SF KICK), SIDECHAIN (Live's sidechain input), GRID (volume shape).
RATE	1/2, 1/4, 1/8 default 1/4	GRID: 1/2, 1/4, 1/8.
CURVE	0 % to 100 % default 50 %	GRID: shape of the recovery.
MIX	0 % to 100 % default 100 %	Amount.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF ENERGY

AUDIO EFFECT

Group energy. Replaces the DJ filter on groups: one bipolar knob, centre neutral, that adds energy as well as taking it away.



USING IT

— One per group (drums, bass, music, voice) with PROFILE to match.

— Down: low-pass, space and softness. Up: high-pass with a resonant edge, drive, 1/16 echo and chop.

— With SLOW on, an 8-bar high-pass build happens by itself.

REFERENCE

NAME	SOUND
PROFILE	DRUMS (high-pass side thins the kick moderately), BASS (strong low-pass side, gentle high-pass), MUSIC (both sides full), VOICE, MASTER (gentle both ways).

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
ENERGY M	LP 100 to HP 100 default OFF	Bipolar. Centre = neutral. Left = down, right = up.
PROFILE	DRUMS, BASS, MUSIC, VOICE, MASTER default MUSIC	How far each side travels for this kind of group.
RES	0 % to 100 % default 30 %	Resonant edge (stronger on the up side).
DRIVE AMT	0 % to 100 % default 40 %	Drive added on the up side (2x oversampled, from clean at centre).
SPACE AMT	0 % to 100 % default 40 %	Reverb added on the down side. It rings out when the knob comes back past centre.
ECHO AMT	0 % to 100 % default 30 %	1/16 echo added on the up side.
GATE AMT	0 % to 100 % default 0 %	1/16 chop added on the up side.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

10 MASTER AND MIX

SF ISOLATOR

AUDIO EFFECT

3-band DJ kill EQ (Linkwitz-Riley, phase-aligned).



USING IT

— Kill the lows before a drop, bring them back on the one.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
LOW	0 % to 100 % default 75 %	0 = kill, 75 % = 0 dB, 100 % = +6 dB.
MID	0 % to 100 % default 75 %	Same for mids.
HIGH	0 % to 100 % default 75 %	Same for highs.
LOW KILL	ON / OFF default OFF	Kill the low band.
MID KILL	ON / OFF default OFF	Kill the mid band.
HIGH KILL	ON / OFF default OFF	Kill the high band.
XOVER LO	60 Hz to 500 Hz default 200 Hz	Low/mid crossover.
XOVER HI	1.00 kHz to 8.00 kHz default 2.50 kHz	Mid/high crossover.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF BUILD

AUDIO EFFECT

One knob for a whole-mix build: high-pass sweep, noise riser, reverb wash and width. DROP lands on the next bar line, with an optional silent GAP before it, the riser's TAIL ringing on, and DROP DUCK clearing the way for the first kicks.



USING IT

— Turn BUILD up over the build. Hold DROP: the drop lands exactly on the next bar line. The display shows DROP ON BAR, then GAP during the gap.

— GAP silences everything through BUILD for that long before the bar line: in the show template the whole LIVE BUS, returns included (BRIDGE and DECK play on). Pressed later than that, the gap starts on the next 1/16.

— TAIL: the riser's noise and the wash ring on for that long after the build stops (where the gap starts, or on the bar with GAP OFF), fading out. Shorter than the GAP, it dies away in the silence; longer, it carries into the drop, DROP DUCK lower. CUT: gone in 5 ms.

— DROP DUCK: for DUCK TIME after the drop, every SF part with Pocket dips this far under each kick, whatever its POCKET DEPTH, so the first kicks land on top of everything that comes back. It starts with the first kick on the bar line (a roll into the drop doesn't duck): full depth for the first half of DUCK TIME, easing out over the second. The parts: BASS, LEAD, PAD, VOICE, RUMBLE, the wet signal of DELAY and REVERB, DUCK, and TAP and STRIP unless their ROLE is KICK, PERC or DECK. Not KICK, PERC, STRIKE, DECK or tracks without an SF plugin. It needs SF KICK reporting (KICK REPORTING on SF CONTROL). 0 dB: off.

— After a drop the build stays off until BUILD is back below 5 % (display: RETURN KNOB), so you can bring the knob down without re-building. An Arrangement loop that wraps while a drop waits lands it

on the wrap; looping back over a drop doesn't duck again.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
BUILD M	0 % to 100 % default 0 %	The build amount.
DROP	HOLD	Arms the drop for the next bar line.
NOISE	0 % to 100 % default 40 %	Noise riser level.
SPACE	0 % to 100 % default 40 %	Reverb wash.
WIDTH	0 % to 100 % default 40 %	Stereo widening.
HP MAX	200 Hz to 4.00 kHz default 1.20 kHz	How high the high-pass goes at full build.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.
GAP	OFF, 1/16, 1/8, 1/4, 1/2, 1 BAR default OFF	Silence before the drop's bar line: OFF, 1/16, 1/8, 1/4, 1/2 or 1 BAR.

SETUP

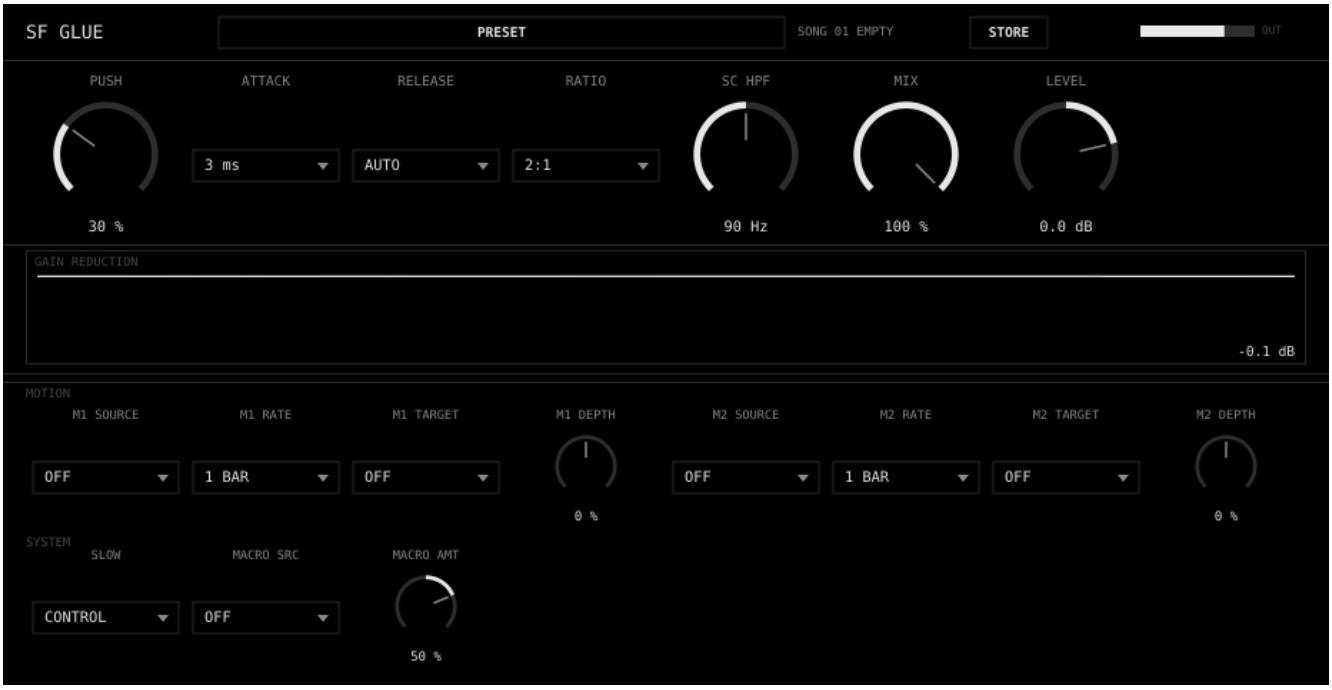
CONTROL	RANGE / DEFAULT	WHAT IT DOES
TAIL	CUT, 1/16, 1/8, 1/4, 1/2, 1 BAR, 2 BARS default 1/8	How long the noise and the wash ring on after the build stops, fading out: CUT (5 ms), 1/16 to 2 BARS.
DROP DUCK	0.0 dB to 12.0 dB default 6.0 dB	How far every SF part with Pocket dips under each kick after the drop, and how much lower the tail runs into it. 0 = off.
DUCK TIME	1/4, 1/2, 1 BAR, 2 BARS, 4 BARS default 1 BAR	How long the drop duck lasts after the bar line: full depth for the first half, easing out over the second.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF GLUE

AUDIO EFFECT

Bus compressor with one PUSH knob (threshold and makeup together). 6 dB soft knee.



USING IT

— PUSH 20-40 % for glue; more for audible pump.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
PUSH M	0 % to 100 % default 30 %	Compression and makeup together.
ATTACK	0.1 ms, 0.3 ms, 1 ms, 3 ms, 10 ms, 30 ms default 3 ms	0.1 to 30 ms.
RELEASE	0.1 s, 0.3 s, 0.6 s, 1.2 s, AUTO default AUTO	0.1 to 1.2 s or AUTO (dual-stage).
RATIO	2:1, 4:1, 10:1 default 2:1	2:1, 4:1, 10:1.
SC HPF	20 Hz to 300 Hz default 90 Hz	High-pass on the detector so the kick doesn't drive it.
MIX	0 % to 100 % default 100 %	Parallel compression.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF TONE

AUDIO EFFECT

Master voicing EQ: five broad moves on one row (low shelf, mud cut, presence, air, tilt). The ranges stop before damage. Runs at 2x, so the air shelf keeps its shape up to 20 kHz.



USING IT

— After GLUE, before LOWEND and LIMIT.

— MUD 1-3 dB cleans a dense mix without thinning it. LOW and AIR at +1-2 dB open up a big system. TILT darkens or brightens everything in one move.

— For the room or PA itself use SF ROOM EQ (stored per venue, after the recording point); TONE is the sound of the set.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
LOW	-6.0 dB to 6.0 dB default 0.0 dB	Low shelf at 70 Hz, +/-6 dB.
MUD	0.0 dB to 6.0 dB default 0.0 dB	Broad cut at 280 Hz, 0 to 6 dB.
PRESENCE	-4.0 dB to 4.0 dB default 0.0 dB	Bell at 3.2 kHz, +/-4 dB.
AIR	-6.0 dB to 6.0 dB default 0.0 dB	High shelf at 12 kHz, +/-6 dB.
TILT	-3.0 dB to 3.0 dB default 0.0 dB	Tilts the whole spectrum around 900 Hz: right brighter, left darker (+/-3 dB at the ends).
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF LOWEND

AUDIO EFFECT

Low-end safety: mono below a frequency, sub clip, rumble cut, width above.



USING IT

— Last-but-one on the master, before LIMIT.

PERFORMANCE ROW

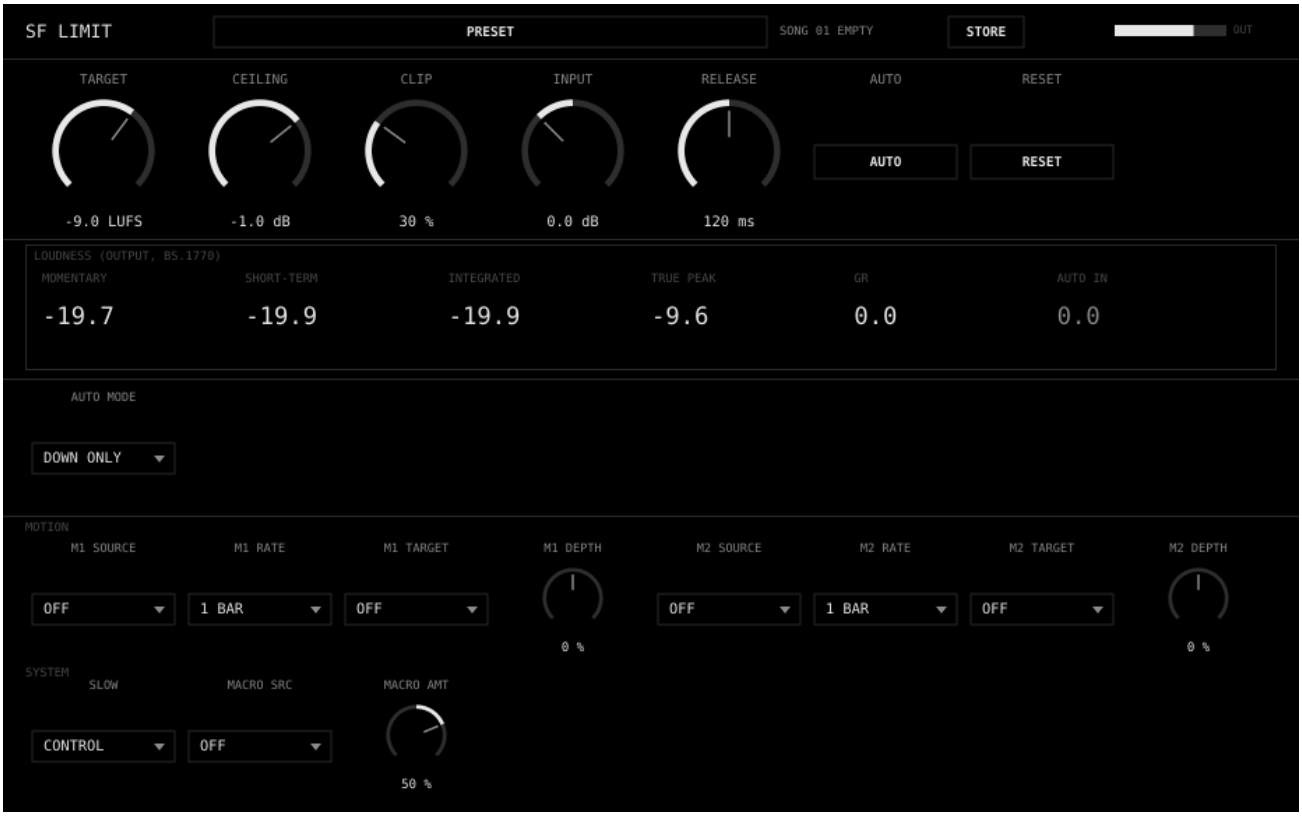
CONTROL	RANGE / DEFAULT	WHAT IT DOES
MONO FREQ	40 Hz to 300 Hz default 120 Hz	Everything below this is mono.
SUB CLIP	0 % to 100 % default 20 %	Soft clip on the mono low band (louder sub, same peak), 2x oversampled.
RUMBLE CUT	10 Hz to 45 Hz default 25 Hz	24 dB/oct high-pass for DC and rumble.
WIDTH	0.00 x to 2.00 x default 1.00 x	Stereo width above MONO FREQ.
LOW TRIM	-12.0 dB to 6.0 dB default 0.0 dB	Level of the low band.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF LIMIT

AUDIO EFFECT LATENCY 176 SAMPLES AT 48 KHZ

Clipper and true-peak lookahead limiter with a BS.1770 loudness meter. The clipper runs at 4x through linear-phase oversampling; the limiter detects true peaks at 8x and never lets the output past CEILING. AUTO trims the input so short-term loudness sits at TARGET.



USING IT

— Last on the master. CEILING -1 dB for venues and releases.

— AUTO keeps the loudest moments of the last minute at TARGET, moving the input by at most 1 dB per second, so a whole set sits at TARGET without anyone riding the master. A breakdown doesn't count as the set getting quieter: AUTO holds through it, and the drop comes back where it was. AUTO MODE DOWN ONLY (the default) never lifts above unity; UP AND DOWN also lifts a quieter set (0.5 dB per second), but never a breakdown.

— CLIP takes the first few dB off the peaks transparently (kick transients) so the limiter works less.

— Recording a set: RESET at the start; INTEGRATED then reads the loudness of the whole set.

— Latency 176 samples at 48 kHz, compensated by Live.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
TARGET	-16.0 LUFS to -5.0 LUFS default -9.0 LUFS	Loudness AUTO aims for (LUFS short-term).
CEILING	-3.0 dB to -0.1 dB default -1.0 dB	True-peak ceiling (dBTP). The output never exceeds it, including between samples.
CLIP	0 % to 100 % default 30 %	Soft clip before the limiter, 4x oversampled, linear phase. The knee starts up to 6 dB below CEILING.
INPUT	-12.0 dB to 24.0 dB default 0.0 dB	Input gain.
RELEASE	20 ms to 500 ms default 120 ms	Limiter release. Under sustained limiting a slower envelope holds the gain, so dense passages don't pump.
AUTO	ON / OFF default OFF	Automatic input trim (+/-12 dB) that keeps the loudest short-term loudness of the last minute at TARGET. It moves up only after 3 s of music within 6 dB of TARGET, so it never lifts a breakdown. Measured on the raw input, so it settles without overshoot.
RESET	HOLD	Resets the integrated loudness and true-peak hold.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
AUTO MODE	DOWN ONLY, UP AND DOWN default DOWN ONLY	DOWN ONLY (default): AUTO never lifts above unity, it only stops the set running above TARGET. UP AND DOWN: it also lifts a quieter set.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF ROOM EQ

AUDIO EFFECT

Cut-only correction for the room or PA, stored per venue (16 venues). Not part of songs.



USING IT

— Put it after the recording point: on an audio track PA with Audio From = Resampling (the Main output), routed to the PA outputs (see Setting up Live). The set is recorded on its own track from the same signal, without the venue's correction.

— Find the boomy and harsh frequencies at soundcheck, cut them, STORE VENUE.

— Next time at that venue, choose it from VENUE.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
CUT 1	0.0 dB to 18.0 dB default 0.0 dB	Cut at band 1 (0 to 18 dB).
CUT 2	0.0 dB to 18.0 dB default 0.0 dB	Cut at band 2 (0 to 18 dB).
CUT 3	0.0 dB to 18.0 dB default 0.0 dB	Cut at band 3 (0 to 18 dB).
CUT 4	0.0 dB to 18.0 dB default 0.0 dB	Cut at band 4 (0 to 18 dB).
CUT 5	0.0 dB to 18.0 dB default 0.0 dB	Cut at band 5 (0 to 18 dB).
CUT 6	0.0 dB to 18.0 dB default 0.0 dB	Cut at band 6 (0 to 18 dB).
Q	0.50 to 8.00 default 2.00	Width of all six bands.
VENUE	VENUE 1, VENUE 2, VENUE 3, VENUE 4, VENUE 5, VENUE 6 ... (16) default VENUE 1	Recalls a stored venue.

SETUP

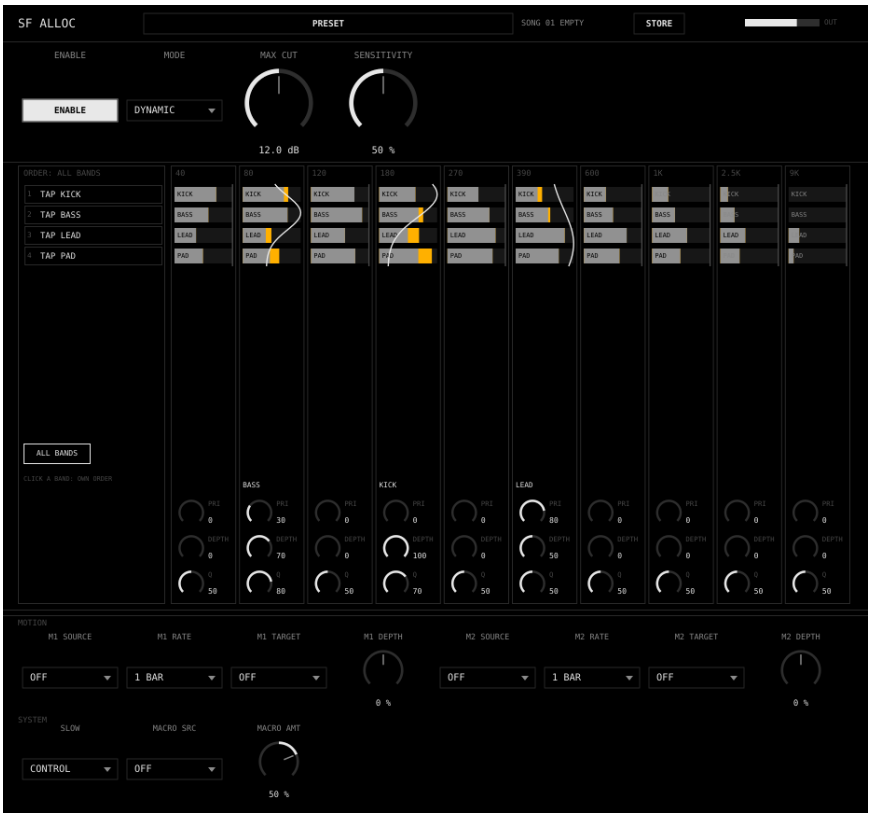
CONTROL	RANGE / DEFAULT	WHAT IT DOES
FREQ 1	30 Hz to 16.00 kHz default 60 Hz	Frequency of band 1.
FREQ 2	30 Hz to 16.00 kHz default 120 Hz	Frequency of band 2.
FREQ 3	30 Hz to 16.00 kHz default 250 Hz	Frequency of band 3.
FREQ 4	30 Hz to 16.00 kHz default 500 Hz	Frequency of band 4.
FREQ 5	30 Hz to 16.00 kHz default 2.00 kHz	Frequency of band 5.
FREQ 6	30 Hz to 16.00 kHz default 6.00 kHz	Frequency of band 6.
STORE VENUE	HOLD	Stores the current bands into the selected venue.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF ALLOC

AUDIO EFFECT

The mix assigner. You put the sources in order; each of the ten bands has a priority curve over that order, set with three knobs, and the display shows exactly what each source keeps and loses in that band. Pass-through audio: put it on the master.



USING IT

— Every SF instrument, SF STRIP, SF TAP and SF DECK appears in the SOURCES list by its Live track name (VST3) or its role (PERC, PERC 2 ...).

— Drag the list into order, top = most important. This is the ORDER: ALL BANDS. It is saved with the set and never changes on its own; new sources are added at the bottom. The order follows each plugin, not its name: reopening the set or renaming a track keeps every source's place, and a duplicated track joins at the bottom.

— Any band can have its own order: click the band (its frame turns white, the list shows ORDER: 180 HZ), drag, and the band is marked OWN. LINK TO ALL puts it back on the global order. Click the band again or ALL BANDS to return to the global list.

— In each band: PRIORITY moves the curve's peak up and down the list, DEPTH sets how hard everything away from the peak is cut (0 = nothing), Q sets the width (narrow: one source owns the band; wide: neighbours share).

— Each band shows its sources in its own order, named in the lanes: grey = what plays, amber = what ALLOC takes away. The name above the knobs is the source the peak sits on.

— MODE DYNAMIC cuts a source only while a more favoured source is actually playing in that band, so when the bass stops the pad gets 180 Hz back within a fraction of a second. STATIC cuts all the time,

like a fixed EQ.

— Knobs are drawn in the window; the 30 band knobs are not meant for hardware, but all are automatable in Live.

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
ENABLE	ON / OFF default ON	Off = no cuts anywhere.
MODE	DYNAMIC, STATIC default DYNAMIC	DYNAMIC (cut only while a more favoured source plays in the band) or STATIC (always).
MAX CUT	0.0 dB to 24.0 dB default 12.0 dB	The largest cut any source can get (DEPTH 100 %, far from the peak).
SENSITIVITY	0 % to 100 % default 50 %	DYNAMIC: how quiet a favoured source can be and still count as playing.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
PRIORITY 1	0 % to 100 % default 0 %	Band 1: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 2	0 % to 100 % default 0 %	Band 2: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 3	0 % to 100 % default 0 %	Band 3: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 4	0 % to 100 % default 0 %	Band 4: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 5	0 % to 100 % default 0 %	Band 5: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 6	0 % to 100 % default 0 %	Band 6: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 7	0 % to 100 % default 0 %	Band 7: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 8	0 % to 100 % default 0 %	Band 8: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 9	0 % to 100 % default 0 %	Band 9: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 10	0 % to 100 % default 0 %	Band 10: where the priority peak sits in the list (0 = top, 100 = bottom).
DEPTH 1	0 % to 100 % default 0 %	Band 1: how hard sources away from the peak are cut.
DEPTH 2	0 % to 100 % default 0 %	Band 2: how hard sources away from the peak are cut.
DEPTH 3	0 % to 100 % default 0 %	Band 3: how hard sources away from the peak are cut.
DEPTH 4	0 % to 100 % default 0 %	Band 4: how hard sources away from the peak are cut.
DEPTH 5	0 % to 100 % default 0 %	Band 5: how hard sources away from the peak are cut.
DEPTH 6	0 % to 100 % default 0 %	Band 6: how hard sources away from the peak are cut.
DEPTH 7	0 % to 100 % default 0 %	Band 7: how hard sources away from the peak are cut.
DEPTH 8	0 % to 100 % default 0 %	Band 8: how hard sources away from the peak are cut.
DEPTH 9	0 % to 100 % default 0 %	Band 9: how hard sources away from the peak are cut.
DEPTH 10	0 % to 100 % default 0 %	Band 10: how hard sources away from the peak are cut.
Q 1	0 % to 100 % default 50 %	Band 1: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 2	0 % to 100 % default 50 %	Band 2: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 3	0 % to 100 % default 50 %	Band 3: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 4	0 % to 100 % default 50 %	Band 4: width of the peak (low = wide, sources share; high = narrow, one owns the band).

CONTROL	RANGE / DEFAULT	WHAT IT DOES
Q 5	0 % to 100 % default 50 %	Band 5: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 6	0 % to 100 % default 50 %	Band 6: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 7	0 % to 100 % default 50 %	Band 7: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 8	0 % to 100 % default 50 %	Band 8: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 9	0 % to 100 % default 50 %	Band 9: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 10	0 % to 100 % default 50 %	Band 10: width of the peak (low = wide, sources share; high = narrow, one owns the band).

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF TAP

AUDIO EFFECT LATENCY 512 SAMPLES AT 48 KHZ

Puts any track (stems, hardware, other plugins) into Pocket and ALLOC. Adds no sound of its own.



USING IT

— One on each stem track from Live's stem separation, with ROLE set (stem drums = PERC, stem bass = BASS ...).

PERFORMANCE ROW

CONTROL	RANGE / DEFAULT	WHAT IT DOES
ROLE	KICK, BASS, VOICE, LEAD, PERC, RUMBLE ... (9) default OTHER	Priority in ALLOC (KICK highest ... OTHER lowest) and the name shown.
ALLOC	ON / OFF default ON	Take part in ALLOC.
POCKET DEPTH	0 % to 100 % default 0 %	Duck this track around SF KICK.
POCKET SHAPE	0 % to 100 % default 30 %	Pocket shape.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

11 APPENDIX

MIDI NOTE MAP

PLUGIN	NOTES	DOES
SF KICK, PERC	any	Plays the sound (KICK: KEY TRACK on makes C1 = TUNE).
SF VOICE	C3, C#3, D3 ...	Slice 1, 2, 3 ... of the phrase. Below C3: the whole phrase.
SF STRIKE	any	Strikes the object (KEY TRACK on: the note sets the pitch, C3 = TUNE).
SF DECK	C1-D#1	Deck A hot cues 1-4 (jump on the next beat).
SF DECK	C2-D#2	Deck B hot cues 1-4.
SF CONTROL	Program Change ch 16	Song 01-16 (channel set by PC CHANNEL).
SF LEAD / BASS ACID	velocity > ~108	Accent. Overlapping notes slide.

WHERE THINGS ARE SAVED

— Songs, samples, ALLOC choices, MOTION random seeds, DECK tracklist and venue EQs: inside the Live set (plugin state).

— User presets: Music/SF Live Suite/Presets/<PLUGIN>, one file per preset.

— DECK track log: Documents/SF Live Suite/tracklog.txt.

— DECK tracks are referenced by file path, not embedded: keep the files where they are.

WHEN SOMETHING ISN'T RIGHT

SYMPTOM	CHECK
SF SHOW.als: a plugin is missing	The suite isn't installed, or VST3 is off (Settings > Plug-Ins). Install or turn VST3 on, then open the set again.
An instrument doesn't play	MIDI From: SF SHOW.als starts every MIDI track on No Input; set each to its ONE port and channel (chapter 7), never All Ins, where every instrument plays every part.
Nothing ducks	SF CONTROL: BUS SHARED and KICK REPORTING. The kick must be an SF KICK (or use SF DUCK with SOURCE SIDECHAIN).
Pump early or late	Live buffer size 512 or less. Then trim POCKET OFFSET on SF CONTROL by ear.
Songs don't change	PC ENABLE on, PC CHANNEL matches the OXI. The VST3 needs Live 12.2 or later; otherwise use the AU.
E16 rings don't follow	Remote on for E16 Port 2's output in Live's MIDI preferences; map with Live's MIDI Map.
An E16 press does nothing	An SF SHOW file from before 3 Nov is loaded (it had no presses): load the current one. A toggle that flips back: Remote is on for two E16 ports, so the note arrives twice. Port 2 only.
An E16 knob jumps when touched	A shared knob holding an old value: controllers/e16 README, check 2. Or Auto Transmit is on: off in the scene and globally.
ALLOC does nothing	ENABLE on, DEPTH above 0 on the band, the instruments' ALLOC toggles on. In DYNAMIC, the favoured source must be playing in that band.
DECK row says MISSING	The file moved. Drop it again.
A knob shows GLIDE	SLOW ALL or the plugin's SLOW is on: the knob is travelling to where you set it.
A knob moves on its own	MOTION: check M1 / M2 TARGET in the plugin's MOTION row.
Header shows FAULT (red)	The plugin caught an invalid value or a runaway and reset itself to silence. It carries on working. If it happens again, note what was playing: it is a bug.
Something won't stop	PANIC on SF CONTROL: every SF plugin fades out, clears and stays silent while held. Live's own clips and devices aren't SF: mute the PA track for those.
Header shows PANIC (red)	PANIC is on at SF CONTROL: held in its window, or flipped on by its key or button. A key flips it back.
CONTROL shows LOCAL ONLY	The plugins come from different builds. Rebuild and install the whole suite.
A return doubles the dry signal	DRY off on SF DELAY, SF REVERB or SF DENSITY on a return track.
Pitch is wrong after a song change	KEY on SF CONTROL is stored per song. KEY FOLLOW off on an instrument ignores it.
Levels changed after the update	Since Nov 3 every model is calibrated to its role's target (chapter 5). Re-balance the faders once.

E16 ADDRESS MAP (SF SHOW)

Every address of the SF SHOW scene once, by channel: the Live target, the mapping's range, and where the knob is on the E16. SF SHOW.als has them all mapped; this list is for checking it, and for mapping a set of your own (Cmd-M). The same list is controllers/e16/sf-show-map.csv.

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
1	CC 20	01 KICK > SF KICK > TUNE	absolute	KICK 9
1	CC 21	01 KICK > SF KICK > DECAY	absolute	KICK 10
1	CC 22	01 KICK > SF KICK > PUNCH	absolute	KICK 11
1	CC 23	01 KICK > SF KICK > CLICK	absolute	KICK 12
1	CC 24	01 KICK > SF KICK > DRIVE	absolute	KICK 15
1	CC 25	01 KICK > SF KICK > FILTER	absolute	KICK 14
1	CC 26	01 KICK > SF KICK > DENSITY	absolute	KICK 13
1	CC 27	01 KICK > SF KICK > LEVEL	absolute	KICK 16
1	CC 100	01 KICK > mixer > volume	absolute, Max 0 dB	every page, encoder 5
2	CC 20	11 BASS > SF BASS > CUTOFF	absolute	BASS 9
2	CC 21	11 BASS > SF BASS > RES	absolute	BASS 10
2	CC 22	11 BASS > SF BASS > ENV AMT	absolute	BASS 11
2	CC 23	11 BASS > SF BASS > GATE	absolute	BASS 12
2	CC 24	11 BASS > SF BASS > DENSITY	absolute	BASS 13
2	CC 25	11 BASS > SF BASS > FILTER	absolute	BASS 14
2	CC 26	11 BASS > SF BASS > DRIVE	absolute	BASS 15
2	CC 27	11 BASS > SF BASS > LEVEL	absolute	BASS 16
3	CC 20	12 RUMBLE > SF RUMBLE > SIZE	absolute	RUMBLE 9
3	CC 21	12 RUMBLE > SF RUMBLE > DARK	absolute	RUMBLE 10
3	CC 22	12 RUMBLE > SF RUMBLE > GRIT	absolute	RUMBLE 11
3	CC 23	12 RUMBLE > SF RUMBLE > PUMP	absolute	RUMBLE 15
3	CC 24	12 RUMBLE > SF RUMBLE > GATE	absolute	RUMBLE 12
3	CC 25	12 RUMBLE > SF RUMBLE > DENSITY	absolute	RUMBLE 13
3	CC 26	12 RUMBLE > SF RUMBLE > FILTER	absolute	RUMBLE 14
3	CC 27	12 RUMBLE > SF RUMBLE > LEVEL	absolute	RUMBLE 16
4	CC 21	21 CLOSED > SF PERC > DECAY	absolute	HATS 10
4	CC 22	21 CLOSED > SF PERC > TONE	absolute	HATS 11
4	CC 23	21 CLOSED > SF PERC > DENSITY	absolute	HATS 9
5	CC 21	22 OPEN > SF PERC > DECAY	absolute	HATS 13
5	CC 22	22 OPEN > SF PERC > TONE	absolute	HATS 14
5	CC 23	22 OPEN > SF PERC > DENSITY	absolute	HATS 12
6	CC 21	23 CLAP > SF PERC > DECAY	absolute	HATS 16
6	CC 23	23 CLAP > SF PERC > DENSITY	absolute	HATS 15
7	CC 20	24 STRIKE > SF STRIKE > POWER	absolute	STRIKE 9
7	CC 21	24 STRIKE > SF STRIKE > DISTANCE	absolute	STRIKE 10
7	CC 22	24 STRIKE > SF STRIKE > SIZE	absolute	STRIKE 11
7	CC 23	24 STRIKE > SF STRIKE > MATERIAL	absolute	STRIKE 15
7	CC 24	24 STRIKE > SF STRIKE > DECAY	absolute	STRIKE 12
7	CC 25	24 STRIKE > SF STRIKE > DENSITY	absolute	STRIKE 13

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
7	CC 26	24 STRIKE > SF STRIKE > FILTER	absolute	STRIKE 14
7	CC 27	24 STRIKE > SF STRIKE > LEVEL	absolute	STRIKE 16
8	CC 20	31 ACID > SF LEAD > CUTOFF	absolute	PERFORM 9, ACID 9
8	CC 21	31 ACID > SF LEAD > RES	absolute	ACID 10
8	CC 22	31 ACID > SF LEAD > ENV AMT	absolute	ACID 11
8	CC 23	31 ACID > SF LEAD > GATE	absolute	ACID 12
8	CC 24	31 ACID > SF LEAD > DENSITY	absolute	ACID 13
8	CC 25	31 ACID > SF LEAD > FILTER	absolute	ACID 14
8	CC 26	31 ACID > SF LEAD > DRIVE	absolute	ACID 15
8	CC 27	31 ACID > SF LEAD > LEVEL	absolute	ACID 16
9	CC 20	32 STAB > SF LEAD > CUTOFF	absolute	STAB 9
9	CC 21	32 STAB > SF LEAD > RES	absolute	STAB 10
9	CC 22	32 STAB > SF LEAD > ENV AMT	absolute	STAB 11
9	CC 23	32 STAB > SF LEAD > GATE	absolute	STAB 12
9	CC 24	32 STAB > SF LEAD > DENSITY	absolute	STAB 13
9	CC 25	32 STAB > SF LEAD > FILTER	absolute	STAB 14
9	CC 27	32 STAB > SF LEAD > LEVEL	absolute	STAB 16
9	CC 39	32 STAB > SF LEAD > SPACE	absolute	STAB 15
9	CC 101	32 STAB > mixer > send A	absolute	PERFORM 10
10	CC 20	33 TEXTURE > SF PAD > CUTOFF	absolute	TEXTURE 9
10	CC 23	33 TEXTURE > SF PAD > GATE	absolute	TEXTURE 12
10	CC 24	33 TEXTURE > SF PAD > DENSITY	absolute	TEXTURE 13
10	CC 25	33 TEXTURE > SF PAD > FILTER	absolute	TEXTURE 14
10	CC 27	33 TEXTURE > SF PAD > LEVEL	absolute	TEXTURE 16
10	CC 30	33 TEXTURE > SF PAD > ATTACK	absolute	TEXTURE 15
10	CC 34	33 TEXTURE > SF PAD > DETUNE	absolute	TEXTURE 11
10	CC 35	33 TEXTURE > SF PAD > COLOR	absolute	PERFORM 11, TEXTURE 10
11	CC 20	34 VOICE > SF VOICE > SIGNAL	absolute	VOICE 9
11	CC 21	34 VOICE > SF VOICE > TONE	absolute	VOICE 10
11	CC 22	34 VOICE > SF VOICE > GATE	absolute	VOICE 12
11	CC 23	34 VOICE > SF VOICE > DENSITY	absolute	VOICE 13
11	CC 24	34 VOICE > SF VOICE > FILTER	absolute	PERFORM 12, VOICE 14
11	CC 25	34 VOICE > SF VOICE > ECHO	absolute	VOICE 11
11	CC 26	34 VOICE > SF VOICE > DRIVE	absolute	VOICE 15
11	CC 27	34 VOICE > SF VOICE > LEVEL	absolute	VOICE 16
12	CC 20	A DUB > SF DELAY > TIME	absolute	SPACE 9
12	CC 21	A DUB > SF DELAY > FEEDBACK	absolute, 0-70 % at first	SPACE 10
12	CC 22	A DUB > SF DELAY > TONE	absolute	SPACE 11
12	CC 23	A DUB > SF DELAY > DRIVE	absolute	SPACE 12
12	CC 24	A DUB > SF DELAY > THROW	press: hold	SPACE 9
12	CC 100	A DUB > mixer > volume	absolute, Max 0 dB	PERFORM 13
12	NOTE 25 (C#0)	A DUB > SF DELAY > FREEZE	press: toggle	SPACE 10

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
13	CC 20	B SPACE > SF REVERB > SIZE	absolute	SPACE 13
13	CC 21	B SPACE > SF REVERB > DECAY	absolute	SPACE 14
13	CC 22	B SPACE > SF REVERB > DARK	absolute	SPACE 15
13	CC 24	B SPACE > SF REVERB > DUCK	absolute	SPACE 16
13	CC 26	B SPACE > SF REVERB > THROW	press: hold	SPACE 13
13	CC 100	B SPACE > mixer > volume	absolute, Max 0 dB	PERFORM 14
13	NOTE 25 (C#0)	B SPACE > SF REVERB > FREEZE	press: toggle	SPACE 14
14	CC 20	10 LOW > SF ENERGY > ENERGY	absolute	every page, encoder 1
14	CC 30	20 DRUMS > SF ENERGY > ENERGY	absolute	every page, encoder 2
14	CC 40	30 MUSIC > SF ENERGY > ENERGY	absolute	every page, encoder 3
14	CC 50	20 DRUMS > SF REPEAT > ROLL	press: hold	MACROS 15
14	CC 51	20 DRUMS > SF REPEAT > LENGTH	absolute	MACROS 15
14	CC 53	20 DRUMS > SF REPEAT > PITCH	absolute	MACROS 16
14	CC 100	10 LOW > mixer > volume	absolute, Max 0 dB	every page, encoder 6
14	CC 101	20 DRUMS > mixer > volume	absolute, Max 0 dB	every page, encoder 7
14	CC 102	30 MUSIC > mixer > volume	absolute, Max 0 dB	every page, encoder 8
15	CC 20	90 LIVE BUS > SF BUILD > BUILD	absolute	every page, encoder 4
15	CC 21	90 LIVE BUS > SF BUILD > DROP	press: hold	every page, encoder 4
15	CC 100	90 LIVE BUS > mixer > volume	absolute, Max 0 dB	PERFORM 15
15	CC 101	40 BRIDGE > mixer > volume	absolute, Max 0 dB	PERFORM 16
16	CC 20	00 CONTROL > SF CONTROL > MACRO 1	absolute	MACROS 9
16	CC 21	00 CONTROL > SF CONTROL > MACRO 2	absolute	MACROS 10
16	CC 22	00 CONTROL > SF CONTROL > MACRO 3	absolute	MACROS 11
16	CC 23	00 CONTROL > SF CONTROL > MACRO 4	absolute	MACROS 12
16	CC 24	00 CONTROL > SF CONTROL > MACRO 5	absolute	MACROS 13
16	CC 25	00 CONTROL > SF CONTROL > MACRO 6	absolute	MACROS 14
16	CC 34	00 CONTROL > SF CONTROL > SLOW ALL	absolute	every page, encoder 1, second destination